

Design and Specification of a Multi-Component Metabolic Subnetwork for the Endothelial Glycocalyx

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1. Introduction: Modeling the Glycocalyx in Endothelial Dysfunction

1.1. The Biological System: A Structurally Complex Sensory Organelle

The endothelial glycocalyx is a dense, carbohydrate-rich layer coating the luminal surface of all blood vessels. It is not a simple passive barrier but a highly dynamic sensory organelle. It acts as the primary interface between the flowing blood and the endothelial cell, playing critical roles in mechanotransduction, vascular permeability, and regulating inflammation and coagulation.

1.2. Pathophysiological Relevance in SHINE: Shedding and Compositional Shifts

In pathological states like SHock-Induced Endotheliopathy (SHINE), the glycocalyx is a primary target of injury. Inflammatory enzymes (e.g., proteases, heparanases) cause massive degradation and "shedding" of the glycocalyx components from the cell surface. This process has two devastating consequences: 1) it destroys the primary vascular barrier, leading to edema, and 2) it can trigger a pro-inflammatory shift in the *composition* of the glycocalyx the cell attempts to resynthesize, further exacerbating the disease state.

1.3. The Modeling Objective: From a Simple Concept to a Dynamic Subnetwork

The objective of this project was to construct a detailed, mechanistically accurate metabolic subnetwork capable of simulating the synthesis, dynamic remodeling, and pathological turnover of the major glycocalyx components. This required moving beyond a simplistic representation to a multi-pathway model that could capture the complex compositional changes and degradation patterns observed in SHINE.

2. Phase 1: Defining the Model's Structure and Topology

2.1. Initial Consideration: A Subsystem within the Cell Membrane

The initial and most straightforward approach was to model all glycocalyx components and reactions as a "subsystem" within the existing Cell Membrane [m] compartment. This was a valid starting point, as the glycocalyx is anchored to the cell membrane.

2.2. The Case for a Dedicated Compartment: Handling *In Situ* Remodeling

A critical turning point in our design process occurred when we decided to include the dynamic, *in situ* enzymatic remodeling of the glycocalyx (e.g., surface sulfatase and heparanase activity). Placing these complex reactions and their unique intermediate metabolites (e.g., Edited_Proteoglycan, Truncated_Proteoglycan) within the general [m] compartment would create significant ambiguity and clutter. To maintain organizational clarity and accurately represent the glycocalyx as a distinct biochemical environment, the decision was made to create a new, dedicated compartment.

2.3. Final Topological Design: The "Glycocalyx Layer [gl]" Compartment

The final, superior strategy is to use a dedicated "Glycocalyx Layer [gl]" compartment. This conceptual space represents the functional layer at the cell surface that is tethered to the membrane. It provides a clean, unambiguous location for all surface-level modification and degradation

reactions, making the model more robust, readable, and extensible. The synthesis factory remains the Golgi [g], and the final delivery of finished products is to this [gl] compartment.

3. Phase 2: Defining the Biochemical Constituents of the Glycocalyx

3.1. Hierarchical Structure of the Glycocalyx

A detailed analysis of the glycocalyx composition revealed a clear hierarchy, which forms the basis for our multi-pathway model.

- **1. Proteoglycans** (Core Protein + GAG Chains)
 - **1.1. Core Proteins** (e.g., Syndecan-1, Versican-1, Aggrecan)
 - 1.1.1. Membrane-Associated Core Proteins
 - **1.1.1.1. Syndecan-1**
 - **1.1.1.2. Syndecan-2**
 - **1.1.1.3. Syndecan-4**
 - **1.1.1.4. Glypsican-1**
 - **1.1.1.5. NG2**
 - **1.1.1.6. Betaglycan**
 - 1.1.2. Secreted & Extracellular Matrix Core Proteins
 - **1.1.2.1. Perlecan**
 - **1.1.2.2. Versican**
 - **1.1.2.3. Biglycan**
 - **1.1.2.4. Decorin**
 - **1.1.2.5. Lumican**
 - **1.1.2.6. Fibromodulin**
 - **1.1.2.7. Serglycin**
 - **1.1.2.8. Aggrecan**
 - **1.2. Attached GAG Chains**
 - **1.2.1. Heparan Sulfate (HS)**
 - **1.2.2. Chondroitin Sulfate (CS)**
 - **1.2.3. Dermatan Sulfate (DS)**
 - **1.2.4. Keratan Sulfate (KS)**
- **2. Hyaluronan (HA)** (A free-floating, non-sulfated GAG chain)

Our model will address each of these major branches with a dedicated set of reactions.

3.2. Core Proteins

3.2.1 Membrane-Associated Core Proteins

- **Syndecan-1:** Synthesis of the Syndecan-1 core protein is a key feature of endothelial cells, where it is a primary component of the glycocalyx that is shed from the cell surface upon inflammatory activation.
- **Syndecan-2:** This core protein is a major syndecan synthesized by human microvascular endothelial cells, and its expression is regulated by growth factors and the surrounding extracellular matrix.
- **Syndecan-4:** Human umbilical vein endothelial cells (HUVECs) express the Syndecan-4 core protein, and its synthesis is upregulated in response to inflammatory cytokines like TNF- α .
- **Glypican-1:** Endothelial cells synthesize the core protein Glypican-1, which localizes to the cell surface and plays a crucial role in the mechanotransduction of blood flow and the production of nitric oxide.
- **NG2:** The NG2 core protein is synthesized by endothelial cells during angiogenesis, where it is involved in endothelial cell migration and the formation of new blood vessels.
- **Betaglycan:** As the Type III TGF- β receptor, the Betaglycan core protein is expressed on endothelial cells and is essential for mediating the anti-migratory effects of TGF- β 2.

3.2.2 Secreted & Extracellular Matrix Core Proteins

- **Perlecan:** Biochemical studies have confirmed that the Perlecan core protein is a major proteoglycan synthesized and secreted by human brain microvascular endothelial cells, forming a key part of the vascular basement membrane.
- **Versican:** Endothelial cells in culture are known to synthesize and secrete the large core protein Versican, which accumulates in the subendothelial matrix.
- **Biglycan:** The synthesis and secretion of the Biglycan core protein by human endothelial cells are significantly induced by the growth factor TGF- β 1.
- **Decorin:** Bovine aortic endothelial cells have been shown to synthesize the Decorin core protein, with its production being regulated by factors such as TGF- β .
- **Lumican:** The Lumican core protein has been identified as a product of endothelial cells that acts as a novel anti-angiogenic factor, capable of inhibiting endothelial cell migration and proliferation.
- **Fibromodulin:** The core protein Fibromodulin is expressed in the endothelium of coronary arteries, with increased expression observed in atherosclerotic plaques.
- **Serglycin:** In addition to hematopoietic cells, human endothelial cells synthesize the Serglycin core protein and constitutively secrete it as a proteoglycan exclusively decorated with chondroitin sulfate chains.
- **Aggrecan:** While Aggrecan is generally absent from healthy blood vessels, it can accumulate in the vessel wall during the development of diseases like atherosclerosis.

3.3 Glycosaminoglycans (GAGs) in the Endothelial Environment

The primary GAGs synthesized by human endothelial cells and attached to core proteins are:

- **Heparan Sulfate (HS):** The most abundant GAG in the endothelial glycocalyx, making up 50-90% of the total GAG content. It is fundamentally important for vascular homeostasis, acting as a mechanosensor for blood flow and binding a vast array of signaling molecules.
- **Chondroitin Sulfate (CS):** The second most common GAG in the endothelial environment. While less abundant than heparan sulfate in the glycocalyx itself, it plays significant roles, particularly in inflammation and vascular remodeling.
- **Hyaluronan (HA):** Unique among GAGs, hyaluronan is not covalently attached to a core protein to form a proteoglycan. Instead, this massive, non-sulfated polysaccharide is synthesized at the cell membrane and remains associated with the glycocalyx, often by binding to receptors like CD44. It forms a structural backbone to which secreted proteoglycans like versican can attach.
- **Keratan Sulfate (KS):** While primarily known for its role in cartilage, keratan sulfate is also present in the vascular system. Its synthesis by endothelial cells is less characterized than HS or CS, but it has been identified on the endothelial surface.

3.4 GAG Attachment to Specific Core Proteins

Below is a list detailing which GAGs are attached to each core protein synthesized by human endothelial cells.

- **Membrane-Associated Core Proteins (Glycocalyx Proper)**
 - **Syndecan-1:** Primarily decorated with Heparan Sulfate chains, but can also carry Chondroitin Sulfate chains. This dual GAG attachment allows it to interact with a wide variety of ligands.
 - **Syndecan-2:** Primarily carries Heparan Sulfate chains.
 - **Syndecan-4:** Almost exclusively decorated with Heparan Sulfate chains.
 - **Glypican-1:** Carries Heparan Sulfate chains. It is a major heparan sulfate proteoglycan of the endothelial cell surface.
 - **NG2 (CSPG4):** As its name "Chondroitin Sulfate Proteoglycan 4" implies, this core protein is predominantly modified with Chondroitin Sulfate chains.
 - **Betaglycan (TGF- β RIII):** Can be modified with both Heparan Sulfate and Chondroitin Sulfate chains.
- **Secreted & Extracellular Matrix Core Proteins (Sub-endothelial Environment):** While not strictly part of the apical glycocalyx that interfaces with blood, these proteoglycans are synthesized by endothelial cells and are crucial to the basement membrane and subendothelial matrix, directly influencing endothelial function.
 - **Perlecan:** A very large proteoglycan that primarily carries Heparan Sulfate chains, making it a cornerstone of the basement membrane's structure and filtering capacity.

- Versican: Exclusively decorated with Chondroitin Sulfate chains. It assembles into the matrix by binding to hyaluronan.
- Biglycan: Typically carries one or two Chondroitin Sulfate or Dermatan Sulfate chains (dermatan sulfate is a modified form of chondroitin sulfate).
- Decorin: Carries a single Chondroitin Sulfate or Dermatan Sulfate chain.
- Lumican: Primarily modified with Keratan Sulfate chains.
- Fibromodulin: Also carries Keratan Sulfate chains.
- Serglycin: Research on serglycin secreted by human endothelial cells shows it is exclusively decorated with Chondroitin Sulfate chains.
- Aggrecan: When synthesized by vascular smooth muscle cells in disease states, this core protein is heavily decorated with both Chondroitin Sulfate and Keratan Sulfate chains.

Core Protein - GAG Chain Association Matrix

Core Protein	Heparan Sulfate (HS)	Chondroitin / Dermatan Sulfate (CS/DS)	Keratan Sulfate (KS)
Membrane-Associated			
Syndecan-1	1	1	0
Syndecan-2	1	0	0
Syndecan-4	1	0	0
Glypican-1	1	0	0
NG2 (CSPG4)	0	1	0
Betaglycan (TGF- β RIII)	1	1	0
Secreted & Extracellular Matrix			
Perlecan	1	0	0
Versican	0	1	0
Biglycan	0	1	0
Decorin	0	1	0
Lumican	0	0	1
Fibromodulin	0	0	1
Serglycin	0	1	0
Aggrecan	0	1	1

4. Phase 3: Design and Specification of the Metabolic Pathways

4.1. Definitive Modeling Strategy: A Dual-Pathway Approach for Proteoglycan Trafficking

The endothelial cell synthesizes a wide array of proteoglycans, but these molecules have fundamentally different biological fates. A sophisticated metabolic model must capture this diversity. Based on our analysis, we have designed a dual-pathway approach that uses distinct model topologies to represent the unique life cycles of **Membrane-Associated Proteoglycans** and **Secreted & Extracellular Matrix (ECM) Proteoglycans**.

4.1.1. Pathway for Membrane-Associated Core Proteins (The "Anchor and Unfurl" Model)

4.1.1.1. Biological Fate and Rationale

This pathway is designed for proteoglycans whose primary function is to be an integral part of the cell surface, acting as receptors, co-receptors, and the very foundation of the apical glycocalyx layer.

- **Examples:** Syndecan-1, -2, -4; Glypican-1; NG2; Betaglycan.
- **Mechanism of Attachment:** These proteins are physically tethered to the cell membrane. Syndecans possess a **transmembrane domain**, a specific alpha-helical segment of the protein that passes directly through the lipid bilayer. Glypicans are attached via a **GPI lipid anchor**. In both cases, they are permanently affixed to the cell surface until they are turned over.
- **Function:** They serve as the direct interface between the cell and the extracellular environment. They bind signaling molecules, transmit mechanical forces from blood flow to the intracellular cytoskeleton, and organize the structure of the glycocalyx. Their fate is tied to the cell that made them.

4.1.1.2. Modeling Pathway: Golgi [g] -> Vesicle [v] -> Membrane [m] -> (Unfurl) -> Membrane (Anchor) [m] + Glycocalix (Ectodomain) [gl] -> Extracellular Space [e]

This multi-step pathway was meticulously designed to be a high-fidelity representation of this biological process.

- **Golgi [g] -> Vesicle [v] (Packaging):** This step represents the finished proteoglycan being sorted and packaged into secretory vesicles at the trans-Golgi network. This is a standard step in the secretory pathway.
- **Vesicle [v] -> Cell Membrane [m] (Delivery and Anchoring):** This is the key delivery event. The model simulates the vesicle fusing with the plasma membrane. At this moment, the proteoglycan cargo is delivered, and its transmembrane domain becomes physically embedded within the lipid bilayer. The entire molecule is now conceptually located *in* the [m] compartment as a single, anchored entity (e.g., Full_Syndecan1_HS_Anchored[m]).
- **(Unfurl) -> Anchor[m] + Ectodomain[gl] (Separation of Domains):** This is the crucial modeling abstraction that resolves the paradox of a molecule existing in two places at once. This reaction represents the molecule settling into its functional state. It consumes the single anchored entity from the membrane and produces two distinct, linked metabolites:
 - **Syndecan1_Anchor[m]:** This metabolite represents the portion of the protein (transmembrane and cytoplasmic domains) that remains physically *within* the cell membrane compartment. Its presence explicitly and visibly connects the glycocalyx to the cell membrane in the model.
 - **Functional_HS_Ectodomain[gl]:** This metabolite represents the large, GAG-chain-rich portion that "unfurls" and projects away from the cell surface, creating the functional glycocalyx layer. It resides in our dedicated [gl] compartment.
- This pathway creates an **explicit, stoichiometric link** between the [m] and [gl] compartments via the "Unfurling" reaction. It correctly models that the fate of the anchor

and the ectodomain can be different (e.g., the ectodomain is shed while the anchor is later degraded intracellularly), providing a level of detail that a simpler model could not achieve.

4.1.2. Pathway for Secreted & Extracellular Matrix Core Proteins (The "Direct Secretion" Model)

4.1.2.1. Biological Fate and Rationale

This pathway is designed for proteoglycans that are synthesized by the endothelial cell but are not destined to be part of its own apical glycocalyx. Instead, they are exported to function in the surrounding environment.

- **Examples:** Versican, Perlecan, Biglycan, Decorin, Serglycin.
- **Mechanism of Release:** These proteins lack a transmembrane domain or a permanent anchor. After being packaged into vesicles, they are released from the cell entirely via **exocytosis**. They become soluble components of the subendothelial space, the basement membrane, or the general extracellular matrix (ECM).
- **Function:** Their role is to build and organize the tissue architecture *outside* the cell. Versican, for example, binds to hyaluronan to form massive, hydrated aggregates that give the ECM its gel-like consistency. Perlecan is a cornerstone of the basement membrane, providing a structural scaffold and a charge-selective filter. Their fate is to become part of the tissue, not part of the cell that made them.

4.1.2.2. Modeling Pathway: Golgi [g] -> Vesicle [v] -> Glycocalyx [gl] -> Extracellular Space [e]

This pathway was designed to model the process of secretion and subsequent association with the cell surface before final deposition.

- **Golgi [g] -> Vesicle [v] (Packaging):** This step is identical to the membrane-associated pathway.
- **Vesicle [v] -> Glycocalyx Layer [gl] (Initial Secretion and Association):** This is the key delivery step for this class. The vesicle fuses with the plasma membrane, but instead of anchoring its cargo, it **releases the soluble proteoglycan** into the immediate pericellular environment. The most accurate place to model this initial release is the [gl] compartment, representing the space immediately adjacent to the cell surface. Here, the secreted proteoglycan (e.g., Functional_Versican[gl]) can interact with other glycocalyx components, such as binding to Hyaluronan.
- **Glycocalyx [gl] -> Extracellular Space [e] (Matrix Deposition / Diffusion):** This final step represents the molecule leaving the immediate cell-surface layer and becoming a stable component of the broader extracellular environment. This could be a reaction like `R_Matrix_Deposition: Functional_Versican[gl] --> Matrix_Versican[e]`. The turnover and shedding reactions (like the MAR reactions) would then act on these metabolites in their final compartments.
- This pathway models the **complete export** of the molecule from the cell. By bypassing the [m] compartment, it accurately reflects that these proteins are not anchored. By including the intermediate [gl] step, it provides a more nuanced and accurate picture than a simple `g -> e`

transport, allowing for the modeling of important cell-surface association events (like Versican binding to Hyaluronan) before the molecule is incorporated into the deeper matrix.

4.2 Considerations before modeling

4.2.1. Core proteins

The aa sequence of the proteins described in the table below will be used to determine the stoichiometry of the core protein synthesis reactions

Core Protein Name	UniProt Accession No.	Gene Name	Number of Amino Acids
Membrane-Associated			
Syndecan-1	P18827	<i>SDC1</i>	310
Syndecan-2	P34741	<i>SDC2</i>	201
Syndecan-4	P31431	<i>SDC4</i>	198
Glypican-1	P35052	<i>GPC1</i>	558
NG2	Q6UVK1	<i>CSPG4</i>	2322
Betaglycan	P37173	<i>TGFBR3</i>	849
Secreted & Extracellular Matrix			
Perlecan	P98160	<i>HSPG2</i>	4391
Versican	P13611	<i>VCAN</i>	3409 (Isoform V0)
Biglycan	P21810	<i>BGN</i>	368
Decorin	P07585	<i>DCN</i>	359
Lumican	P51884	<i>LUM</i>	338
Fibromodulin	Q06828	<i>FMOD</i>	376
Serglycin	P10124	<i>SRGN</i>	158
Aggrecan	P16112	<i>ACAN</i>	2415

4.2.2. GAG

4.2.2. Stoichiometric Parameters for GAG Degradation Pathways (The "N-to-1" Model)

A central feature of the glycocalyx subnetwork is its ability to model the pathological degradation of long, structural Glycosaminoglycan (GAG) chains into multiple, smaller, bioactive fragments. This is represented by an "N-to-1" stoichiometric model, where one large polymer is cleaved into 'N' smaller ones. The value of 'N' for each GAG type is not arbitrary but is calculated based on literature-derived values for the average native chain length and the average size of the resulting bioactive fragments.

4.2.2.1 Heparan Sulfate (HS)

Parameter to Justify: N = 10 for the reaction R_HS_Heparanase_Cleavage.

This value is calculated as $N = (\text{Average native HS chain length}) / (\text{Average bioactive fragment size})$.

- **Justification for Native HS Chain Length (~100 disaccharide units):**
 - **Source:** Iozzo, R. V., & Schaefer, L. (2015). Proteoglycan form and function: A comprehensive nomenclature of proteoglycans. *Matrix Biology*, 42, 11–55.

- **Rationale:** The GAG chains on cell-surface proteoglycans like syndecans are heterogeneous, typically ranging from 70 to 250 saccharide units (35 to 125 disaccharides). A representative average of **100 disaccharide units** (approx. 50 kDa) is a standard and scientifically accepted value used for modeling these chains, representing a balance between shorter and longer forms.
- **Justification for Degrading Enzyme and Bioactive Fragment Size (~10 disaccharide units):**
 - **Source:** Vlodavsky, I., et al. (2007). Mammalian heparanase: gene cloning, expression and function in health and disease. *Glycoconjugate journal*, 24(1), 43-55.
 - **Rationale:** The key degrading enzyme is **Heparanase (HPSE)**, an endoglycosidase massively upregulated in inflammation. It cleaves HS chains into substantial fragments, as very small pieces are not biologically active. The resulting fragments are typically 5-10 kDa, corresponding to roughly **10 disaccharide units**. These 10-unit fragments are the minimum size required to effectively bind and present growth factors, a key pro-inflammatory function.
- **Final Calculation:** $N = 100 \text{ disaccharides} / 10 \text{ disaccharides} = 10$.

4.2.2.2 Chondroitin Sulfate (CS)

Parameter to Justify: $N = 12$ for the Chondroitin Sulfate shedding reactions.

- **Justification for Native CS Chain Length (~100 disaccharide units):**
 - **Source:** Iozzo, R. V., & Schaefer, L. (2015).
 - **Rationale:** Similar to HS, the CS chains on proteoglycans like Syndecan-1 and Versican are long polymers. Their size is comparable to HS chains, with typical lengths of up to 250 saccharides. We will use a representative average of **100 disaccharide units** (~50 kDa) for consistency and as a physiologically relevant length.
- **Justification for Degrading Enzyme and Bioactive Fragment Size (~8 disaccharide units):**
 - **Source:** Wight, T. N., & Potter-Perigo, S. (2011). The role of proteoglycans in tissue inflammation. *Glycobiology*, 21(1), 10-21.
 - **Rationale:** The primary degrading enzymes for CS are **Hyaluronidases** (e.g., HYAL1, HYAL2) and certain proteases like MMPs. The resulting fragments have been shown to act as "danger signals," binding to receptors like TLR4 to promote inflammation. Studies on the inflammatory activity of CS fragments indicate that the bioactive units are small oligosaccharides, typically ranging from 6 to 10 disaccharides in length. We will use a representative average size of **8 disaccharide units** for these pro-inflammatory fragments.
- **Final Calculation:** $N = 100 \text{ disaccharides} / 8 \text{ disaccharides} = 12.5$. For a stoichiometric model, this is rounded to the nearest integer, **N = 12**.

4.2.2.3 Keratan Sulfate (KS)

Parameter to Justify: $N = 5$ for Keratan Sulfate GAG chain cleavage reactions.

- **Biological Context:** To create the most comprehensive model of the glycocalyx, a pathway for Keratan Sulfate (KS) is included. While less abundant on endothelial cells than Heparan or Chondroitin Sulfate, Keratan Sulfate proteoglycans (KSPGs) like **Aggrecan (ACAN)** have a documented presence in the vascular system and may be upregulated in specific pathological conditions. This pathway is modeled as a true parallel life cycle, beginning with the synthesis of its own unique core protein.
- **Justification for Native KS Chain Length (~25 disaccharide units):**
 - **Source:** Caterson, B., & Melrose, J. (2018). Keratan sulfate, a complex glycosaminoglycan with multifaceted roles in immune and inflammatory systems. *Glycobiology*, 28(4), 191–207.
 - **Rationale:** This review details the structure of KS on large proteoglycans like Aggrecan. KS chains are known for their heterogeneity and are shorter than HS/CS chains, typically ranging from 6 to 50 saccharides. A representative average length for a mature KS chain on Aggrecan is approximately **50 saccharides**, which corresponds to **25 disaccharide units** (with a molecular weight of ~10-12 kDa).
- **Justification for Degrading Enzyme and Bioactive Fragment Size (~5 disaccharide units):**
 - **Source:** Funderburgh, J. L. (2000). Keratan sulfate: structure, biosynthesis, and function. *Glycobiology*, 10(10), 951-958. (and supported by Caterson & Melrose, 2018).
 - **Rationale:** KS is degraded by a series of endo- and exoglycosidases, including specific **keratanases**. In inflammatory contexts, the fragments of KS released are not inert; they have been shown to be bioactive, for example, by modulating cytokine activity. The literature on bioactive GAG fragments suggests that oligosaccharides of a certain minimal length are required for receptor engagement. A representative size for a bioactive KS fragment is an oligosaccharide of 8-10 sugar units, which corresponds to approximately **5 disaccharide units**. This size is consistent with fragments large enough to possess specific signaling functions, aligning with our model's goal of representing the conversion of structural components into pro-inflammatory signals.
- **Final Calculation:** $N = 25 \text{ disaccharides} / 5 \text{ disaccharides} = 5$.

4.2.2.4 Dermatan Sulfate (DS)

Parameter to Justify: $N = 12$ for Dermatan Sulfate shedding reactions.

- **Biological Context:** Dermatan Sulfate is a structural isomer of Chondroitin Sulfate, where some of the Glucuronic Acid (GlcA) units are epimerized to Iduronic Acid (IdoA). It is often found on the same core proteins as CS and is functionally significant in vascular biology.
- **Justification for Native DS Chain Length (~100 disaccharide units):**

- **Source:** Iozzo, R. V., & Schaefer, L. (2015).
- **Rationale:** As DS is a modification of CS, its chain length is comparable. We will use the same representative average of **100 disaccharide units**.
- **Justification for Degrading Enzyme and Bioactive Fragment Size (~8 disaccharide units):**
 - **Source:** Wight, T. N., & Potter-Perigo, S. (2011).
 - **Rationale:** DS is also a substrate for Hyaluronidases, and its fragments are also known to be bioactive. Due to its close structural relationship to CS, we will use the same representative bioactive fragment size of **8 disaccharide units**.
- **Final Calculation:** $N = 100 \text{ disaccharides} / 8 \text{ disaccharides} = 12.5$. This is rounded to $N = 12$.

4.2.2.5. Summary Table of "N" Values

GAG Type	Avg. Native Length (Disaccharides)	Avg. Fragment Size (Disaccharides)	Calculated N	Biological Relevance in Endothelium
Heparan Sulfate (HS)	100	10	10	Very High (Primary Component)
Chondroitin Sulfate (CS)	100	8	12	High (Increases in Inflammation)
Dermatan Sulfate (DS)	100	8	12	High (Present with CS)
Keratan Sulfate (KS)	25	5	5	Low (Not a major endothelial GAG)

4.2.3. Others

4.2.3.1. ATP

Parameter to Justify: $\text{ATP Cost} = 4 * (\text{Number of Amino Acids} - 1)$.

- **Source:** Thiele, I., & Palsson, B. Ø. (2010). A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols*, 5(1), 93–121. (and general biochemistry textbooks).
- **Rationale:** This is the standard, universally accepted energy cost for protein translation used in all modern genome-scale models. The cost breaks down as follows for each peptide bond formed:
 1. **Amino Acid Activation:** The attachment of an amino acid to its tRNA molecule consumes two high-energy phosphate bonds ($\text{ATP} \rightarrow \text{AMP} + \text{PPi}$). (**Cost = 2 ATP**).
 2. **Ribosome Translocation:** The movement of the ribosome along the mRNA to the next codon, a step catalyzed by elongation factors, consumes one molecule of GTP ($\text{GTP} \rightarrow \text{GDP} + \text{Pi}$). This is energetically equivalent to one ATP. (**Cost = 1 ATP**).
 3. **tRNA Entry:** The binding of the charged tRNA to the ribosome's A-site also consumes one molecule of GTP. (**Cost = 1 ATP**).
- **Total cost per peptide bond = 2 + 1 + 1 = 4 ATP equivalents.**

- Since a protein with 'X' amino acids has 'X-1' peptide bonds, the total cost is $4 * (X-1)$. Our model uses this exact, well-established calculation for all protein synthesis reactions.

4.2.3.2 H2O

Parameter to Justify: $H_2O \text{ Cost} = (\text{ATP Cost}) - (\text{Number of Amino Acids} - 1)$

- **Source:** Thiele, I., & Palsson, B. Ø. (2010). A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols*, 5(1), 93–121. (and general biochemistry principles).
- **Rationale:** The net stoichiometry of water in a protein synthesis reaction is the result of two opposing processes that happen simultaneously: **water consumption** during energy hydrolysis and **water production** during peptide bond formation. The model's reaction represents the sum of these two events.

Let's break it down:

1. Water is CONSUMED during ATP Hydrolysis:

- The overall energy cost of forming one peptide bond is the hydrolysis of 4 high-energy phosphate bonds, which is represented as the conversion of 4 ATP to 4 ADP and 4 Pi.
- The chemical reaction for the hydrolysis of one ATP molecule is:
 $ATP + H_2O \rightarrow ADP + Pi + H^+$
- Therefore, to pay the energy cost for forming (X-1) peptide bonds, the cell must consume $4 * (X-1)$ molecules of water.
- **Water Consumed = $4 * (X - 1) = \text{ATP Cost}$**

2. Water is PRODUCED during Peptide Bond Formation (Condensation):

- A peptide bond is formed when the carboxyl group of one amino acid links to the amino group of the next.
- This is a **condensation reaction** (also called a dehydration synthesis reaction), which means a molecule of water is released for every bond that is formed.
- The chemical reaction is:
 $\text{AminoAcid}_1 + \text{AminoAcid}_2 \rightarrow \text{Dipeptide} + H_2O$
- For a protein with X amino acids, there are (X-1) peptide bonds formed. Therefore, the cell produces (X-1) molecules of water during this process.
- **Water Produced = $(X - 1)$**

3. Calculating the NET Water Stoichiometry:

- The final stoichiometric coefficient for water in our reaction is the net balance of water consumed versus water produced.
- **Net H2O = (Water Consumed) - (Water Produced)**
- **Net H2O = $[4 * (X - 1)] - [(X - 1)]$**

- This can be simplified to:
Net H₂O = 3 * (X - 1)
- Alternatively, using the term we already defined:
Net H₂O = (ATP Cost) - (Number of Amino Acids - 1)

Example using Syndecan-1 (251 Amino Acids):

- **Number of Amino Acids (X):** 251
- **Number of Peptide Bonds (X-1):** 250
- **ATP Cost:** 4 * 250 = 1000
- **Water Consumed (for ATP hydrolysis):** 1000 H₂O
- **Water Produced (from condensation):** 250 H₂O
- **Net H₂O Stoichiometry (on reactant side):** 1000 - 250 = 750 H₂O

4.3. Membrane-Associated Core Proteins

This section details the metabolic life cycles of proteoglycans that are integral to the cell membrane.

4.3.1. Syndecan-1

4.3.1.1. Heparan Sulfate (HS) Pathway

This subnetwork details the complete life cycle of a Syndecan-1 proteoglycan decorated with a representative Heparan Sulfate GAG chain.

(This block defines the necessary transport of all individual precursors to the specific synthesis sites)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 individual reversible reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_Golgi	udpglcna[c] <=> udpglcna[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_Sulfate_GL	sulfate[e] <=> sulfate[gl]	Glycocalyx Layer [gl]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]

(This block details the complete, explicit synthesis of the Syndecan-1 core protein and a 100-disaccharide Heparan Sulfate chain)

Reaction ID	Reaction Equation	Compartment
R_Synth_SDC1_Core	35 ala-L[er] + 10 arg-L[er] + 5 asn-L[er] + 11 asp-L[er] + 2 cys-L[er] + 13 gln-L[er] + 26 glu-L[er] + 24 gly[er] + 7 his-L[er] + 5 ile-L[er] + 22 leu-L[er] + 11 lys-L[er] + 4 met-L[er] + 3 phe-L[er] + 22 pro-L[er] + 17 ser-L[er] + 24 thr-L[er] + 3 trp-L[er] + 5 tyr-L[er] + 12 val-L[er] + 1000 atp[er] + 750 h2o[er] --> Syndecan1_Core_Protein[er] + 1000 adp[er] + 1000 pi[er] + 1000 h[er]	ER [er]
T_Core_SDC1_ER_Golgi	Syndecan1_Core_Protein[er] --> Syndecan1_Core_Protein[g]	ER [er] -> Golgi [g]
R_HS_Elongation_on_SDC1	Syndecan1_Core_Protein[g] + 100 udpglcnac[g] + 100 udpglca[g] --> HS_Chain_100_on_SDC1_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_HS_Sulfation_on_SDC1	HS_Chain_100_on_SDC1_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_HS_Proteoglycan_on_SDC1[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the refined "Anchor and Unfurl" delivery pathway)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_SDC1_Packaging	Fully_Assembled_HS_Proteoglycan_on_SDC1[g] --> Packaged_HS_on_SDC1[v]	Golgi [g] -> Vesicle [v]
R_HS_on_SDC1_Vesicle_Activation	Packaged_HS_on_SDC1[v] + atp[c] + gtp[c] --> Activated_HS_on_SDC1_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_HS_on_SDC1_Delivery	Activated_HS_on_SDC1_Cargo[v] --> Full_Syndecan1_HS_Anchored[m]	Vesicle [v] -> Cell Membrane [m]

(This block includes the "Unfurling" step, the dual-pathology turnover model, and all necessary demand reactions to ensure the network is unblocked)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_SDC1_Unfurling	Full_Syndecan1_HS_Anchored[m] --> Syndecan1_Anchor[m] + Functional_HS_Ectodomain_on_SDC1[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_HS_on_SDC1_Sulfatase_Editing	Functional_HS_Ectodomain_on_SDC1[gl] + h2o[gl] --> Edited_HS_Ectodomain_on_SDC1[gl] + sulfate[gl] + h[gl]	Glycocalyx [gl]
R_HS_on_SDC1_Proteolytic_Shedding	Syndecan1_Anchor[m] + Functional_HS_Ectodomain_on_SDC1[gl] --> Syndecan1_Stub[m] + Shed_Syndecan1_HS_Ectodomain[e]	Membrane [m] + Glycocalyx [gl] -> [m] + Extracellular [e]
R_HS_on_SDC1_Heparanase_Cleavage	Functional_HS_Ectodomain_on_SDC1[gl] + 9 H2O [gl] -->	Glycocalyx [gl] -> [gl] + Extracellular [e]

	Truncated_HS_Ectodomain_on_SDC1[gl] + 10 Shed_HS_Fragment[e] + 9 H[gl]	
DM_Anchor_SDC1	Syndecan1_Anchor[m] -->	Cell Membrane [m]
DM_Edited_Ectodomain_HS_on_SDC1	Edited_HS_Ectodomain_on_SDC1[gl] -->	Glycocalyx [gl]
DM_Truncated_Ectodomain_HS_on_SDC1	Truncated_HS_Ectodomain_on_SDC1[gl] -->	Glycocalyx [gl]
DM_Stub_SDC1	Syndecan1_Stub[m] -->	Cell Membrane [m]
DM_Shed_Ectodomain_HS_on_SDC1	Shed_Syndecan1_HS_Ectodomain[e] -->	Extracellular [e]
DM_Shed_HS_Fragment	Shed_HS_Fragment[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
Syndecan1_Core_Protein	[er], [g]	The unmodified 251 amino acid protein backbone of Syndecan-1.
HS_Chain_100_on_SDC1_Unsulfated	[g]	The SDC1 core with a 100-disaccharide unsulfated Heparan Sulfate chain.
Fully_Assembled_HS_Proteoglycan_on_SDC1	[g]	The final product in the Golgi: SDC1 with a 100-unit sulfated HS chain.
Packaged_HS_on_SDC1	[v]	The finished SDC1-HS proteoglycan packaged within a transport vesicle.
Activated_HS_on_SDC1_Cargo	[v]	The SDC1-HS vesicle cargo after being "powered up" for transport.
Full_Syndecan1_HS_Anchored	[m]	The complete SDC1-HS proteoglycan, newly delivered and anchored in the membrane.
Syndecan1_Anchor	[m]	The transmembrane and cytoplasmic domains of SDC1, physically in the membrane.
Functional_HS_Ectodomain_on_SDC1	[gl]	The extracellular portion (protein + HS chain) creating the glycocalyx layer.
Edited_HS_Ectodomain_on_SDC1	[gl]	The ectodomain after surface-level enzymatic editing (de-sulfation).
Truncated_HS_Ectodomain_on_SDC1	[gl]	The ectodomain remaining at the surface after heparanase has cleaved its HS chains.
Syndecan1_Stub	[m]	The remaining transmembrane/cytoplasmic protein stub after proteolytic shedding.
Shed_Syndecan1_HS_Ectodomain	[e]	The entire soluble ectodomain (protein + HS chains) released after proteolytic shedding.
Shed_HS_Fragment	[e]	A soluble, 10-disaccharide HS fragment released after heparanase cleavage.

4.3.1.2. Chondroitin Sulfate (CS) Pathway

This subnetwork details the complete life cycle of a Syndecan-1 proteoglycan decorated with a representative Chondroitin Sulfate GAG chain. It uses the detailed, multi-step degradation cascade derived from your provided MAR reactions.

(This block is conceptually identical to the HS pathway and uses the same transporters. It ensures the ER and Golgi are stocked with the necessary precursors for CS synthesis from the cytosol [c])

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 individual reversible reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GalNAc_Golgi	udpgal 鈇[c] <=> udpgal 鈇[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]

*(This block begins with the **shared** Syndecan-1 core protein and then details the now-separate synthesis of the three different CS GAG chains onto it. The stoichiometry for the chain length is now corrected to 100 disaccharides.)*

Reaction ID	Reaction Equation	Compartment
R_Synth_SDC1_Core	35 ala-L[er] + 10 arg-L[er] + 5 asn-L[er] + 11 asp-L[er] + 2 cys-L[er] + 13 gln-L[er] + 26 glu-L[er] + 24 gly[er] + 7 his-L[er] + 5 ile-L[er] + 22 leu-L[er] + 11 lys-L[er] + 4 met-L[er] + 3 phe-L[er] + 22 pro-L[er] + 17 ser-L[er] + 24 thr-L[er] + 3 trp-L[er] + 5 tyr-L[er] + 12 val-L[er] + 1000 atp[er] + 750 h2o[er] --> Syndecan1_Core_Protein[er] + 1000 adp[er] + 1000 pi[er] + 1000 h[er]	ER [er]
T_Core_SDC1_ER_Golgi	Syndecan1_Core_Protein[er] --> Syndecan1_Core_Protein[g]	ER [er] -> Golgi [g]
R_CS_Elongation_on_SDC1	Syndecan1_Core_Protein[g] + 100 udpgal 鈇[g] + 100 udpglca[g] --> CS_Chain_100_on_SDC1_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_CS_Sulfation_A_B_on_SDC1	CS_Chain_100_on_SDC1_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_CSA_B_Proteoglycan_on_SDC1[g] + 100 pap[g]	Golgi [g]
R_CS_Epimerization_and_Sulfation_B_on_SDC1	CS_Chain_100_on_SDC1_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_CSB_Proteoglycan_on_SDC1[g] + 100 pap[g]	Golgi [g]
R_CS_Sulfation_E_on_SDC1	CS_Chain_100_on_SDC1_Unsulfated[g] + 200 paps[g] --> Fully_Assembled_CSE_Proteoglycan_on_SDC1[g] + 200 pap[g]	Golgi [g]

T_Waste_UDP_out udp[g] <=> udp[c] Golgi [g] -> Cytosol [c]

T_Waste_PAP_out pap[g] <=> pap[c] Golgi [g] -> Cytosol [c]

(This block now includes fully separate, parallel, and unambiguous delivery pathways for the distinct CSA, CSB, and CSE proteoglycan types)

Reaction ID	Reaction Equation	Compartments Involved
R_CSA_on_SDC1_Packaging	Fully_Assembled_CSA_Proteoglycan_on_SDC1[g] --> Packaged_CSA_on_SDC1[v]	Golgi [g] -> Vesicle [v]
R_CSA_on_SDC1_Vesicle_Activation	Packaged_CSA_on_SDC1[v] + atp[c] + gtp[c] --> Activated_CSA_on_SDC1_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_CSA_on_SDC1_Delivery	Activated_CSA_on_SDC1_Cargo[v] --> Full_Syndecan1_CSA_Anchored[m]	Vesicle [v] -> Cell Membrane [m]
R_CSB_on_SDC1_Packaging	Fully_Assembled_CSB_Proteoglycan_on_SDC1[g] --> Packaged_CSB_on_SDC1[v]	Golgi [g] -> Vesicle [v]
R_CSB_on_SDC1_Vesicle_Activation	Packaged_CSB_on_SDC1[v] + atp[c] + gtp[c] --> Activated_CSB_on_SDC1_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_CSB_on_SDC1_Delivery	Activated_CSB_on_SDC1_Cargo[v] --> Full_Syndecan1_CSB_Anchored[m]	Vesicle [v] -> Cell Membrane [m]
R_CSE_on_SDC1_Packaging	Fully_Assembled_CSE_Proteoglycan_on_SDC1[g] --> Packaged_CSE_on_SDC1[v]	Golgi [g] -> Vesicle [v]
R_CSE_on_SDC1_Vesicle_Activation	Packaged_CSE_on_SDC1[v] + atp[c] + gtp[c] --> Activated_CSE_on_SDC1_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_CSE_on_SDC1_Delivery	Activated_CSE_on_SDC1_Cargo[v] --> Full_Syndecan1_CSE_Anchored[m]	Vesicle [v] -> Cell Membrane [m]

(This block now correctly applies the N=12 "N-to-1" degradation model for Chondroitin Sulfate chains, using the superior "Anchor and Unfurl" topology)

Reaction ID (Model)	Reaction ID (Original)	Reaction Equation	Compartments Involved
R_CSA_on_SDC1_Unfurling	N/A	Full_Syndecan1_CSA_Anchored[m] --> Syndecan1_Anchor[m] + Functional_CSA_Ectodomain_on_SDC1[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_CSB_on_SDC1_Unfurling	N/A	Full_Syndecan1_CSB_Anchored[m] --> Syndecan1_Anchor[m] + Functional_CSB_Ectodomain_on_SDC1[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_CSE_on_SDC1_Unfurling	N/A	Full_Syndecan1_CSE_Anchored[m] --> Syndecan1_Anchor[m] + Functional_CSE_Ectodomain_on_SDC1[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_CS_Fragmentation_A	Based on MAR	Functional_CSA_Ectodomain_on_SDC1[gl] + 11 H2O[gl] -->	Glycocalyx [gl] -> Membrane [m] +

	reactions	Syndecan1_Stub_from_CSA[m] + 12 Shed_CS_Fragment_A[e] + 11 H[gl]	Extracellular [e]
R_CS_Fragmentation_B	Based on MAR reactions	Functional_CSB_Ectodomain_on_SDC1[gl] + 11 H2O[gl] --> Syndecan1_Stub_from_CSB[m] + 12 Shed_CS_Fragment_B[e] + 11 H[gl]	Glycocalyx [gl] -> Membrane [m] + Extracellular [e]
R_CS_Fragmentation_E	Based on MAR reactions	Functional_CSE_Ectodomain_on_SDC1[gl] + 11 H2O[gl] --> Syndecan1_Stub_from_CSE[m] + 12 Shed_CS_Fragment_E[e] + 11 H[gl]	Glycocalyx [gl] -> Membrane [m] + Extracellular [e]
DM_Anchor_SDC1	N/A	Syndecan1_Anchor[m] -->	Cell Membrane [m]
DM_Stub_SDC1_A	N/A	Syndecan1_Stub_from_CSA[m] -->	Cell Membrane [m]
DM_Stub_SDC1_B	N/A	Syndecan1_Stub_from_CSB[m] -->	Cell Membrane [m]
DM_Stub_SDC1_E	N/A	Syndecan1_Stub_from_CSE[m] -->	Cell Membrane [m]
DM_Clearance_E	N/A	Shed_CS_Fragment_A[e] → Shed_CS_Fragment_B[e] → Shed_CS_Fragment_E[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
CS_Chain_100_on_SDC1_Unsulfated	[g]	The SDC1 core with a 100-disaccharide unsulfated Chondroitin Sulfate chain.
Fully_Assembled_CSA_B_Proteoglycan_on_SDC1	[g]	The final SDC1 proteoglycan in the Golgi with a CS A/B type sulfation pattern.
Fully_Assembled_CSE_Proteoglycan_on_SDC1	[g]	The final SDC1 proteoglycan in the Golgi with a CS E type sulfation pattern.
Packaged_..., Activated_...	[v]	Vesicular transport intermediates for the different CS types on SDC1.
Full_Syndecan1_CSA/B/E_Anchored	[m]	The complete, anchored SDC1 proteoglycans with different CS types in the membrane.
Functional_CSA/B/E_Ectodomain_on_SDC1	[gl]	The functional ectodomains (protein + CS chain) for each CS type in the glycocalyx layer.
Shed_CS_Fragment_...	[e]	The various soluble fragments released into the extracellular space after shedding.
GalNAc	[e]	N-Acetyl-D-galactosamine released during degradation.

4.3.2. Syndecan-2

4.3.2.1. Heparan Sulfate (HS) Pathway

This subnetwork details the complete life cycle of a Syndecan-2 proteoglycan decorated with a representative 100-disaccharide Heparan Sulfate GAG chain.

(This block is identical to the one used for Syndecan-1, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_Sulfate_GL	sulfate[e] <=> sulfate[gl]	Glycocalyx Layer [gl]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]

(This block details the synthesis of the unique Syndecan-2 core protein and the attachment of HS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_SDC2_Core	17 ala-L[er] + 10 arg-L[er] + 5 asn-L[er] + 15 asp-L[er] + 2 cys-L[er] + 4 gln-L[er] + 16 glu-L[er] + 11 gly[er] + 4 his-L[er] + 10 ile-L[er] + 21 leu-L[er] + 11 lys-L[er] + 4 met-L[er] + 4 phe-L[er] + 10 pro-L[er] + 18 ser-L[er] + 14 thr-L[er] + 1 trp-L[er] + 6 tyr-L[er] + 12 val-L[er] + 800 atp[er] + 600 h2o[er] --> Syndecan2_Core_Protein[er] + 800 adp[er] + 800 pi[er] + 800 h[er]	ER [er]
T_Core_SDC2_ER_Golgi	Syndecan2_Core_Protein[er] --> Syndecan2_Core_Protein[g]	ER [er] -> Golgi [g]
R_HS_Elongation_on_SDC2	Syndecan2_Core_Protein[g] + 100 udpglcnac[g] + 100 udpglca[g] --> HS_Chain_100_on_SDC2_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_HS_Sulfation_on_SDC2	HS_Chain_100_on_SDC2_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_HS_Proteoglycan_on_SDC2[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the "Anchor and Unfurl" delivery pathway for the new SDC2 proteoglycan)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_SDC2_Packaging	Fully_Assembled_HS_Proteoglycan_on_SDC2[g] --> Packaged_HS_on_SDC2[v]	Golgi [g] -> Vesicle [v]
R_HS_on_SDC2_Vesicle_Activation	Packaged_HS_on_SDC2[v] + atp[c] + gtp[c] --> Activated_HS_on_SDC2_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_HS_on_SDC2_Delivery	Activated_HS_on_SDC2_Cargo[v] --> Full_Syndecan2_HS_Anchored[m]	Vesicle [v] -> Cell Membrane [m]

(This block is parallel to the SDC1-HS pathway, with its own set of metabolites and independent fluxes)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_SDC2_Unfurling	Full_Syndecan2_HS_Anchored[m] --> Syndecan2_Anchor[m] + Functional_HS_Ectodomain_on_SDC2[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_HS_on_SDC2_Sulfatase_Editing	Functional_HS_Ectodomain_on_SDC2[gl] + h2o[gl] --> Edited_HS_Ectodomain_on_SDC2[gl] + sulfate[gl] + h[gl]	Glycocalyx [gl]
R_HS_on_SDC2_Proteolytic_Shedding	Syndecan2_Anchor[m] + Functional_HS_Ectodomain_on_SDC2[gl] --> Syndecan2_Stub[m] + Shed_Syndecan2_HS_Ectodomain[e]	Membrane [m] + Glycocalyx [gl] -> [m] + Extracellular [e]
R_HS_on_SDC2_Heparanase_Cleavage	Functional_HS_Ectodomain_on_SDC2[gl] + 9 H2O[gl] --> Truncated_HS_Ectodomain_on_SDC2[gl] + 10 Shed_HS_Fragment[e] + 9 H[gl]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Anchor_SDC2	Syndecan2_Anchor[m] -->	Cell Membrane [m]
DM_Edited_Ectodomain_HS_on_SDC2	Edited_HS_Ectodomain_on_SDC2[gl] -->	Glycocalyx [gl]
DM_Truncated_Ectodomain_HS_on_SDC2	Truncated_HS_Ectodomain_on_SDC2[gl] -->	Glycocalyx [gl]
DM_Stub_SDC2	Syndecan2_Stub[m] -->	Cell Membrane [m]
DM_Shed_Ectodomain_HS_on_SDC2	Shed_Syndecan2_HS_Ectodomain[e] -->	Extracellular [e]
(Shared Sink)	Shed_HS_Fragment[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
Syndecan2_Core_Protein	[er], [g]	The unmodified 201 amino acid protein backbone of Syndecan-2 (UniProt P34741).
HS_Chain_100_on_SDC2_Unsulfated	[g]	The SDC2 core with a 100-disaccharide unsulfated Heparan Sulfate chain.
Fully_Assembled_HS_Proteoglycan_on_SDC2	[g]	The final product in the Golgi: SDC2 with a 100-unit sulfated HS chain.
Packaged_HS_on_SDC2	[v]	The finished SDC2-HS proteoglycan packaged within a transport vesicle.
Activated_HS_on_SDC2_Cargo	[v]	The SDC2-HS vesicle cargo after being "powered up" for transport.
Full_Syndecan2_HS_Anchored	[m]	The complete SDC2-HS proteoglycan, newly delivered and anchored in the membrane.
Syndecan2_Anchor	[m]	The transmembrane and cytoplasmic domains of SDC2, physically in the membrane.
Functional_HS_Ectodomain_on_SDC2	[gl]	The extracellular portion (protein + HS chain) of SDC2 creating the glycocalyx layer.

Edited_HS_Ectodomain_on_SDC2	[gl]	The SDC2 ectodomain after surface-level enzymatic editing (de-sulfation).
Truncated_HS_Ectodomain_on_SDC2	[gl]	The SDC2 ectodomain remaining after heparanase has cleaved its HS chains.
Syndecan2_Stub	[m]	The remaining transmembrane/cytoplasmic protein stub of SDC2 after proteolytic shedding.
Shed_Syndecan2_HS_Ectodomain	[e]	The entire soluble ectodomain (protein + HS chains) of SDC2 released after proteolytic shedding.
Shed_HS_Fragment	[e]	A soluble, 10-disaccharide HS fragment (shared product with SDC1 pathway).

4.3.3. Syndecan-4

4.3.3.1. Heparan Sulfate (HS) Pathway

This subnetwork details the complete life cycle of a Syndecan-4 proteoglycan decorated with a representative 100-disaccharide Heparan Sulfate GAG chain.

(This block is identical to the one used for the other proteoglycan pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_Sulfate_GL	sulfate[e] <=> sulfate[gl]	Glycocalyx Layer [gl]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]

(This block details the synthesis of the unique Syndecan-4 core protein and the attachment of HS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_SDC4_Core	14 ala-L[er] + 8 arg-L[er] + 4 asn-L[er] + 13 asp-L[er] + 0 cys-L[er] + 4 gln-L[er] + 22 glu-L[er] + 19 gly[er] + 2 his-L[er] + 9 ile-L[er] + 18 leu-L[er] + 11 lys-L[er] + 3 met-L[er] + 6 phe-L[er] + 12 pro-L[er] + 16 ser-L[er] + 8 thr-L[er] + 0 trp-L[er] + 4 tyr-L[er] + 18 val-L[er] + 788 atp[er] + 591 h2o[er] --> Syndecan4_Core_Protein[er] + 788 adp[er] + 788 pi[er] + 788 h[er]	ER [er]
T_Core_SDC4_ER_Golgi	Syndecan4_Core_Protein[er] --> Syndecan4_Core_Protein[g]	ER [er] -> Golgi [g]

R_HS_Elongation_on_SDC4 Syndecan4_Core_Protein[g] + 100 udpglcnac[g] + 100
udpglca[g] --> HS_Chain_100_on_SDC4_Unsulfated[g] + 200 Golgi [g]
udp[g]
R_HS_Sulfation_on_SDC4 HS_Chain_100_on_SDC4_Unsulfated[g] + 100 paps[g] --> Golgi [g]
Fully_Assembled_HS_Proteoglycan_on_SDC4[g] + 100 pap[g]
T_Waste_UDP_out udp[g] <=> udp[c] Golgi [g] ->
Cytosol [c]
T_Waste_PAP_out pap[g] <=> pap[c] Golgi [g] ->
Cytosol [c]
(This block details the "Anchor and Unfurl" delivery pathway for the new SDC4 proteoglycan)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_SDC4_Packaging	Fully_Assembled_HS_Proteoglycan_on_SDC4[g] --> Packaged_HS_on_SDC4[v]	Golgi [g] -> Vesicle [v]
R_HS_on_SDC4_Vesicle_Activation	Packaged_HS_on_SDC4[v] + atp[c] + gtp[c] --> Activated_HS_on_SDC4_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_HS_on_SDC4_Delivery	Activated_HS_on_SDC4_Cargo[v] --> Full_Syndecan4_HS_Anchored[m]	Vesicle [v] -> Cell Membrane [m]

(This block is parallel to the other Syndecan-HS pathways, with its own set of metabolites and independent fluxes)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_SDC4_Unfurling	Full_Syndecan4_HS_Anchored[m] --> Syndecan4_Anchor[m] + Functional_HS_Ectodomain_on_SDC4[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_HS_on_SDC4_Sulfatase_Editing	Functional_HS_Ectodomain_on_SDC4[gl] + h2o[gl] --> Edited_HS_Ectodomain_on_SDC4[gl] + sulfate[gl] + h[gl]	Glycocalyx [gl]
R_HS_on_SDC4_Proteolytic_Shedding	Syndecan4_Anchor[m] + Functional_HS_Ectodomain_on_SDC4[gl] --> Syndecan4_Stub[m] + Shed_Syndecan4_HS_Ectodomain[e]	Membrane [m] + Glycocalyx [gl] -> [m] + Extracellular [e]
R_HS_on_SDC4_Heparanase_Cleavage	Functional_HS_Ectodomain_on_SDC4[gl] + 9 H2O[gl] --> Truncated_HS_Ectodomain_on_SDC4[gl] + 10 Shed_HS_Fragment[e] + 9 H[gl]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Anchor_SDC4	Syndecan4_Anchor[m] -->	Cell Membrane [m]
DM_Edited_Ectodomain_HS_on_SDC4	Edited_HS_Ectodomain_on_SDC4[gl] -->	Glycocalyx [gl]
DM_Truncated_Ectodomain_HS_on_SDC4	Truncated_HS_Ectodomain_on_SDC4[gl] -->	Glycocalyx [gl]
DM_Stub_SDC4	Syndecan4_Stub[m] -->	Cell Membrane [m]
DM_Shed_Ectodomain_HS_on_SDC4	Shed_Syndecan4_HS_Ectodomain[e] -->	Extracellular [e]
(Shared Sink)	Shed_HS_Fragment[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
Syndecan4_Core_Protein	[er], [g]	The unmodified 198 amino acid protein backbone of Syndecan-4 (UniProt P31431).
HS_Chain_100_on_SDC4_Unsulfated	[g]	The SDC4 core with a 100-disaccharide unsulfated Heparan Sulfate chain.
Fully_Assembled_HS_Proteoglycan_on_SDC4	[g]	The final product in the Golgi: SDC4 with a 100-unit sulfated HS chain.
Packaged_HS_on_SDC4	[v]	The finished SDC4-HS proteoglycan packaged within a transport vesicle.
Activated_HS_on_SDC4_Cargo	[v]	The SDC4-HS vesicle cargo after being "powered up" for transport.
Full_Syndecan4_HS_Anchored	[m]	The complete SDC4-HS proteoglycan, newly delivered and anchored in the membrane.
Syndecan4_Anchor	[m]	The transmembrane and cytoplasmic domains of SDC4, physically in the membrane.
Functional_HS_Ectodomain_on_SDC4	[gl]	The extracellular portion (protein + HS chain) of SDC4 creating the glycocalyx layer.
Edited_HS_Ectodomain_on_SDC4	[gl]	The SDC4 ectodomain after surface-level enzymatic editing (de-sulfation).
Truncated_HS_Ectodomain_on_SDC4	[gl]	The SDC4 ectodomain remaining after heparanase has cleaved its HS chains.
Syndecan4_Stub	[m]	The remaining transmembrane/cytoplasmic protein stub of SDC4 after proteolytic shedding.
Shed_Syndecan4_HS_Ectodomain	[e]	The entire soluble ectodomain (protein + HS chains) of SDC4 released after proteolytic shedding.
Shed_HS_Fragment	[e]	A soluble, 10-disaccharide HS fragment (shared product with other HS pathways).

4.3.4. Glypican-1

4.3.4.1. Heparan Sulfate (HS) Pathway

This subnetwork details the complete life cycle of a Glypican-1 proteoglycan, decorated with a representative 100-disaccharide Heparan Sulfate GAG chain and anchored to the membrane via a GPI lipid anchor.

(This block is comprehensive, including precursors for the protein core, the HS chain, and the new GPI anchor)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]

T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
--- New Transporters for GPI Anchor ---		
T_PI_ER	phosphatidylinositol[c] <=> phosphatidylinositol[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_ER	udpglcnac[c] <=> udpglcnac[er]	Endoplasmic Reticulum [er]
T_Dol_P_Man_ER	dolichol-P-mannose[c] <=> dolichol-P-mannose[er]	Endoplasmic Reticulum [er]
T_PE_ER	phosphatidylethanolamine[c] <=> phosphatidylethanolamine[er]	Endoplasmic Reticulum [er]
T_Sulfate_GL	sulfate[e] <=> sulfate[gl]	Glycocalyx Layer [gl]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block includes the synthesis of the unique GPC1 core protein, the parallel synthesis of the GPI anchor, and their final attachment)

Reaction ID	Reaction Equation	Compartment
R_Synth_GPC1_Core	48 ala-L[er] + 33 arg-L[er] + 13 asn-L[er] + 29 asp-L[er] + 14 cys-L[er] + 17 gln-L[er] + 33 glu-L[er] + 33 gly[er] + 5 his-L[er] + 11 ile-L[er] + 54 leu-L[er] + 15 lys-L[er] + 4 met-L[er] + 12 phe-L[er] + 21 pro-L[er] + 33 ser-L[er] + 17 thr-L[er] + 8 trp-L[er] + 9 tyr-L[er] + 26 val-L[er] + 2228 atp[er] + 1671 h2o[er] --> Glypican1_Core_Protein[er] + 2228 adp[er] + 2228 pi[er] + 2228 h[er]	ER [er]
R_Synth_GPI_Anchor	phosphatidylinositol[er] + udpglcnac[er] + 3 dolichol-P-mannose[er] + phosphatidylethanolamine[er] --> GPI_Anchor[er] + udp[er] + 3 dolichol-P[er] + 3 man[er]	ER [er]
R_Attach_GPI_to_GPC1	Glypican1_Core_Protein[er] + GPI_Anchor[er] --> GPC1_with_Anchor[er]	ER [er]
T_Core_GPC1_ER_Golgi	GPC1_with_Anchor[er] --> GPC1_with_Anchor[g]	ER [er] -> Golgi [g]
R_HS_Elongation_on_GPC1	GPC1_with_Anchor[g] + 100 udpglcnac[g] + 100 udpglca[g] --> HS_Chain_100_on_GPC1_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_HS_Sulfation_on_GPC1	HS_Chain_100_on_GPC1_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_HS_Proteoglycan_on_GPC1[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the delivery of the finished, GPI-anchored GPC1 proteoglycan)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_GPC1_Packaging	Fully_Assembled_HS_Proteoglycan_on_GPC1[g] --> Packaged_HS_on_GPC1[v]	Golgi [g] -> Vesicle [v]
R_HS_on_GPC1_Vesicle_Activation	Packaged_HS_on_GPC1[v] + atp[c] + gtp[c] --> Activated_HS_on_GPC1_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_HS_on_GPC1_Delivery	Activated_HS_on_GPC1_Cargo[v] --> Full_Glypican1_HS_Anchored[m]	Vesicle [v] -> Cell Membrane [m]
<i>(This block is parallel to the Syndecan pathways but uses the new GPC1 metabolites)</i>		

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_GPC1_Unfurling	Full_Glypican1_HS_Anchored[m] --> Glypican1_GPI_Anchor[m] + Functional_HS_Ectodomain_on_GPC1[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_HS_on_GPC1_Shedding	Functional_HS_Ectodomain_on_GPC1[gl] --> Shed_Glypican1_HS_Ectodomain[e] <i>(Note: GPC1 shedding is often mediated by lipases like GPLD1 that cleave the GPI anchor itself)</i>	Glycocalyx [gl] -> Extracellular [e]
R_HS_on_GPC1_Heparanase_Cleavage	Functional_HS_Ectodomain_on_GPC1[gl] + 9 H2O[gl] --> Truncated_HS_Ectodomain_on_GPC1[gl] + 10 Shed_HS_Fragment[e] + 9 H[gl]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Anchor_GPC1	Glypican1_GPI_Anchor[m] -->	Cell Membrane [m]
DM_Truncated_Ectodomain_HS_on_GPC1	Truncated_HS_Ectodomain_on_GPC1[gl] -->	Glycocalyx [gl]
DM_Shed_Ectodomain_HS_on_GPC1	Shed_Glypican1_HS_Ectodomain[e] -->	Extracellular [e]
(Shared Sink)	Shed_HS_Fragment[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
GPI_Anchor	[er]	The complete Glycosylphosphatidylinositol lipid anchor.
Glypican1_Core_Protein	[er]	The unmodified 558 amino acid protein backbone of Glypican-1 (UniProt P35052).
GPC1_with_Anchor	[er], [g]	The GPC1 core protein covalently attached to its GPI anchor.
Fully_Assembled_HS_Proteoglycan_on_GPC1	[g]	The final product in the Golgi: GPC1-GPI with a 100-unit sulfated HS chain.
Full_Glypican1_HS_Anchored	[m]	The complete GPC1-HS proteoglycan, newly delivered and anchored in the membrane via its GPI anchor.
Glypican1_GPI_Anchor	[m]	The GPI lipid anchor of GPC1, physically in the membrane.
Functional_HS_Ectodomain_on_GPC1	[gl]	The extracellular protein portion with its HS chain, creating the glycocalyx layer.
Truncated_HS_Ectodomain	[gl]	The GPC1 ectodomain remaining after heparanase has

n_on_GPC1		cleaved its HS chains.
Shed_Glypican1_HS_Ectodomain	[e]	The entire soluble ectodomain (protein + HS chains) of GPC1 released after shedding.

4.3.5. NG2 (CSPG4)

4.3.5.1. Chondroitin Sulfate (CS) and Dermatan Sulfate (DS) Pathway

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GalNAc_Golgi	udpgal 𐄂[c] <=> udpgal 𐄂[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_Sulfate_GL	sulfate[e] <=> sulfate[gl]	Glycocalyx Layer [gl]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

Reaction ID	Reaction Equation	Compartment
R_Synth_NG2_Core	173 ala-L[er] + 104 arg-L[er] + 80 asn-L[er] + 143 asp-L[er] + 22 cys-L[er] + 99 gln-L[er] + 211 glu-L[er] + 193 gly[er] + 41 his-L[er] + 83 ile-L[er] + 228 leu-L[er] + 82 lys-L[er] + 22 met-L[er] + 71 phe-L[er] + 135 pro-L[er] + 189 ser-L[er] + 141 thr-L[er] + 12 trp-L[er] + 45 tyr-L[er] + 148 val-L[er] + 9284 atp[er] + 6963 h2o[er] --> NG2_Core_Protein[er] + 9284 adp[er] + 9284 pi[er] + 9284 h[er]	ER [er]
T_Core_NG2_ER_Golgi	NG2_Core_Protein[er] --> NG2_Core_Protein[g]	ER [er] -> Golgi [g]
R_CS_Elongation_on_NG2	NG2_Core_Protein[g] + 100 udpgal 𐄂[g] + 100 udpglca[g] --> CS_Chain_100_on_NG2_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_CS_Sulfation_A_B_on_NG2	CS_Chain_100_on_NG2_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_CSA_B_Proteoglycan_on_NG2[g] + 100 pap[g]	Golgi [g]
R_CS_Sulfation_E_on_NG2	CS_Chain_100_on_NG2_Unsulfated[g] + 200 paps[g] --> Fully_Assembled_CSE_Proteoglycan_on_NG2[g] + 200 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

Reaction ID	Reaction Equation	Compartments Involved
R_CSA_B_on_NG2_Packaging	Fully_Assembled_CSA_B_Proteoglycan_on_NG2[g] --> Packaged_CSA_B_on_NG2[v]	Golgi [g] -> Vesicle [v]
R_CSA_B_on_NG2_Vesicle_Activation	Packaged_CSA_B_on_NG2[v] + atp[c] + gtp[c] --> Activated_CSA_B_on_NG2_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_CSA_on_NG2_Delivery	Activated_CSA_B_on_NG2_Cargo[v] --> Full_NG2_CSA_Anchored[m]	Vesicle [v] -> Cell Membrane [m]
R_CSB_on_NG2_Delivery	Activated_CSA_B_on_NG2_Cargo[v] --> Full_NG2_CSB_Anchored[m]	Vesicle [v] -> Cell Membrane [m]
R_CSE_on_NG2_Packaging	Fully_Assembled_CSE_Proteoglycan_on_NG2[g] --> Packaged_CSE_on_NG2[v]	Golgi [g] -> Vesicle [v]
R_CSE_on_NG2_Vesicle_Activation	Packaged_CSE_on_NG2[v] + atp[c] + gtp[c] --> Activated_CSE_on_NG2_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_CSE_on_NG2_Delivery	Activated_CSE_on_NG2_Cargo[v] --> Full_NG2_CSE_Anchored[m]	Vesicle [v] -> Cell Membrane [m]

Reaction ID (Model)	Reaction ID (Original)	Reaction Equation	Compartments Involved
R_CSA_on_NG2_Unfurling	N/A	Full_NG2_CSA_Anchored[m] --> NG2_Anchor[m] + Functional_CSA_Ectodomain_on_NG2[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_CSB_on_NG2_Unfurling	N/A	Full_NG2_CSB_Anchored[m] --> NG2_Anchor[m] + Functional_CSB_Ectodomain_on_NG2[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_CSE_on_NG2_Unfurling	N/A	Full_NG2_CSE_Anchored[m] --> NG2_Anchor[m] + Functional_CSE_Ectodomain_on_NG2[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_CS_Shedding_A_on_NG2	Based on MAR16365	Functional_CSA_Ectodomain_on_NG2[gl] + 11 H2O[gl] --> NG2_Stub_from_CSA[m] + 12 Shed_CS_Fragment_A[e] + 11 H[gl]	Glycocalyx [gl] -> Membrane [m] + Extracellular [e]
R_CS_Shedding_B_on_NG2	Based on MAR16377	Functional_CSB_Ectodomain_on_NG2[gl] + 11 H2O[gl] --> NG2_Stub_from_CSB[m] + 12 Shed_CS_Fragment_B[e] + 11 H[gl]	Glycocalyx [gl] -> Membrane [m] + Extracellular [e]
R_CS_Shedding_E_on_NG2	Based on MAR21784	Functional_CSE_Ectodomain_on_NG2[gl] + 11 H2O[gl] --> NG2_Stub_from_CSE[m] + 12 Shed_CS_Fragment_E[e] + 11 H[gl]	Glycocalyx [gl] -> Membrane [m] + Extracellular [e]
DM_Anchor_NG2	N/A	NG2_Anchor[m] →	Cell Membrane [m]

DM_Stub_NG2_A	N/A	NG2_Stub_from_CSA[m] →	Cell Membrane [m]
DM_Stub_NG2_B	N/A	NG2_Stub_from_CSB[m] →	Cell Membrane [m]
DM_Stub_NG2_E	N/A	NG2_Stub_from_CSE[m] →	Cell Membrane [m]
DM_Clearance_E	N/A	Shed_CS_Fragment_A[e] → Shed_CS_Fragment_B[e] → Shed_CS_Fragment_E[e] →	Extracellular [e]

Definition of New Metabolites for this Pathway (Fully Expanded)

Metabolite ID	Compartment	Description
NG2_Core_Protein	[er], [g]	The unmodified 2322 amino acid protein backbone of NG2/CSPG4 (UniProt Q6UVK1).
CS_Chain_100_on_NG2_Unsulfated	[g]	The NG2 core with a 100-disaccharide unsulfated CS chain.
Fully_Assembled_CSA_B_Proteoglycan_on_NG2	[g]	Final NG2 proteoglycan in Golgi with a CS A/B type sulfation.
Fully_Assembled_CSE_Proteoglycan_on_NG2	[g]	Final NG2 proteoglycan in Golgi with a CS E type sulfation.
Packaged_CSA_B_on_NG2, Packaged_CSE_on_NG2	[v]	The finished NG2 proteoglycans packaged within vesicles.
Activated_CSA_B_on_NG2_Cargo, Activated_CSE_on_NG2_Cargo	[v]	The NG2 vesicle cargoes after being "powered up" for transport.
Full_NG2_CSA_Anchored, Full_NG2_CSB_Anchored, Full_NG2_CSE_Anchored	[m]	The complete, anchored NG2 proteoglycans with different CS types in the membrane.
NG2_Anchor	[m]	The transmembrane and cytoplasmic domains of NG2, physically in the membrane.
Functional_CSA_Ectodomain_on_NG2, , ..._CSB_..., ..._CSE_...	[gl]	The functional ectodomains (protein + CS chain) for each CS type in the glycocalyx layer.
NG2_Stub_from_CSA, ..._CSB_..., ..._CSE_...	[m]	The remaining protein stub of NG2 after shedding of the respective ectodomain.
Shed_CS_Fragment_A, Shed_CS_Fragment_B, Shed_CS_Fragment_E	[e]	Soluble fragments of the respective CS types released into the extracellular space.

4.3.6. Betaglycan (TGF-β RIII)

4.3.6.1. Heparan Sulfate (HS) Pathway

This subnetwork details the complete life cycle of a Betaglycan proteoglycan decorated with a representative 100-disaccharide Heparan Sulfate GAG chain.

(This block is identical to the one used for the other proteoglycan pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_Sulfate_GL	sulfate[e] <=> sulfate[gl]	Glycocalyx Layer [gl]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]

(This block details the synthesis of the unique Betaglycan core protein and the attachment of HS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Betaglycan_Core	53 ala-L[er] + 44 arg-L[er] + 37 asn-L[er] + 53 asp-L[er] + 28 cys-L[er] + 31 gln-L[er] + 61 glu-L[er] + 69 gly[er] + 20 his-L[er] + 41 ile-L[er] + 85 leu-L[er] + 37 lys-L[er] + 16 met-L[er] + 28 phe-L[er] + 57 pro-L[er] + 63 ser-L[er] + 46 thr-L[er] + 11 trp-L[er] + 22 tyr-L[er] + 77 val-L[er] + 3392 atp[er] + 2544 h2o[er] --> Betaglycan_Core_Protein[er] + 3392 adp[er] + 3392 pi[er] + 3392 h[er]	ER [er]
T_Core_Betaglycan_ER_Golgi	Betaglycan_Core_Protein[er] --> Betaglycan_Core_Protein[g]	ER [er] -> Golgi [g]
R_HS_Elongation_on_Betaglycan	Betaglycan_Core_Protein[g] + 100 udpglcnac[g] + 100 udpglca[g] --> HS_Chain_100_on_Betaglycan_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_HS_Sulfation_on_Betaglycan	HS_Chain_100_on_Betaglycan_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_HS_Proteoglycan_on_Betaglycan[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the "Anchor and Unfurl" delivery pathway for the new Betaglycan proteoglycan)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_Betaglycan_Packaging	Fully_Assembled_HS_Proteoglycan_on_Betaglycan[g] --> Packaged_HS_on_Betaglycan[v]	Golgi [g] -> Vesicle [v]
R_HS_on_Betaglycan_Vesicle_Activation	Packaged_HS_on_Betaglycan[v] + atp[c] + gtp[c] --> Activated_HS_on_Betaglycan_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]

R_HS_on_Betaglycan_Delivery Activated_HS_on_Betaglycan_Cargo[v] --> Vesicle [v] -> Cell Membrane [m]
Full_Betaglycan_HS_Anchored[m]
(This block is parallel to the Syndecan-HS pathway, with its own set of metabolites and independent fluxes)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_Betaglycan_Unfurling	Full_Betaglycan_HS_Anchored[m] --> Betaglycan_Anchor[m] + Functional_HS_Ectodomain_on_Betaglycan[gl]	Membrane [m] -> [m] + Glycocalyx [gl]
R_HS_on_Betaglycan_Sulfatase_Editing	Functional_HS_Ectodomain_on_Betaglycan[gl] + h2o[gl] --> Edited_HS_Ectodomain_on_Betaglycan[gl] + sulfate[gl] + h[gl]	Glycocalyx [gl]
R_HS_on_Betaglycan_Proteolytic_Shedding	Betaglycan_Anchor[m] + Functional_HS_Ectodomain_on_Betaglycan[gl] --> Betaglycan_Stub[m] + Shed_Betaglycan_HS_Ectodomain[e]	Membrane [m] + Glycocalyx [gl] -> [m] + Extracellular [e]
R_HS_on_Betaglycan_Heparanase_Cleavage	Functional_HS_Ectodomain_on_Betaglycan[gl] + 9 H2O[gl] --> Truncated_HS_Ectodomain_on_Betaglycan[gl] + 10 Shed_HS_Fragment[e] + 9 H[gl]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Anchor_Betaglycan	Betaglycan_Anchor[m] -->	Cell Membrane [m]
DM_Edited_Ectodomain_in_HS_on_Betaglycan	Edited_HS_Ectodomain_on_Betaglycan[gl] -->	Glycocalyx [gl]
DM_Truncated_Ectodomain_HS_on_Betaglycan	Truncated_HS_Ectodomain_on_Betaglycan[gl] -->	Glycocalyx [gl]
DM_Stub_Betaglycan	Betaglycan_Stub[m] -->	Cell Membrane [m]
DM_Shed_Ectodomain_HS_on_Betaglycan	Shed_Betaglycan_HS_Ectodomain[e] -->	Extracellular [e]
(Shared Sink)	Shed_HS_Fragment[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
Betaglycan_Core_Protein	[er], [g]	The unmodified 849 amino acid protein backbone of Betaglycan (UniProt P37173).
HS_Chain_100_on_Betaglycan_Unsulfated	[g]	The Betaglycan core with a 100-disaccharide unsulfated Heparan Sulfate chain.
Fully_Assembled_HS_Proteoglycan_on_Betaglycan	[g]	The final product in the Golgi: Betaglycan with a 100-unit sulfated HS chain.
Packaged_HS_on_Betaglycan	[v]	The finished Betaglycan-HS proteoglycan packaged within a transport vesicle.
Activated_HS_on_Betaglycan_Cargo	[v]	The Betaglycan-HS vesicle cargo after being "powered up" for transport.
Full_Betaglycan_HS_Anchored	[m]	The complete Betaglycan-HS proteoglycan, newly delivered and anchored in the membrane.

Betaglycan_Anchor	[m]	The transmembrane and cytoplasmic domains of Betaglycan, physically in the membrane.
Functional_HS_Ectodomain_on_Betaglycan	[gl]	The extracellular portion (protein + HS chain) of Betaglycan creating the glycocalyx layer.
Edited_HS_Ectodomain_on_Betaglycan	[gl]	The Betaglycan ectodomain after surface-level enzymatic editing (de-sulfation).
Truncated_HS_Ectodomain_on_Betaglycan	[gl]	The Betaglycan ectodomain remaining after heparanase has cleaved its HS chains.
Betaglycan_Stub	[m]	The remaining transmembrane/cytoplasmic protein stub of Betaglycan after proteolytic shedding.
Shed_Betaglycan_HS_Ectodomain	[e]	The entire soluble ectodomain (protein + HS chains) of Betaglycan released after proteolytic shedding.
Shed_HS_Fragment	[e]	A soluble, 10-disaccharide HS fragment (shared product with other HS pathways).

4.4. Secreted & Extracellular Matrix Core Proteins

4.4.1. Perlecan

4.4.1.1. Heparan Sulfate (HS) Pathway

This subnetwork details the complete life cycle of a Perlecan proteoglycan, which is secreted to and degraded within the pericellular environment.

(This block is identical to the one used for the other proteoglycan pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_GL	h[e] <=> h[gl]	Glycocalyx Layer [gl]
T_H2O_GL	h2o[e] <=> h2o[gl]	Glycocalyx Layer [gl]

(This block details the synthesis of the unique Perlecan core protein with the newly verified stoichiometry)

Reaction ID	Reaction Equation	Compartment
R_Synth_Perlecan_Core	333 ala-L[er] + 206 arg-L[er] + 147 asn-L[er] + 250 asp-L[er] + 132 cys-L[er] + 172 gln-L[er] + 392 glu-L[er] + 456 gly[er] + 106 his-L[er] + 139 ile-L[er] + 354 leu-L[er] + 109 lys-L[er] + 40	ER [er]

met-L[er] + 123 phe-L[er] + 327 pro-L[er] + 386 ser-L[er] + 268 thr-L[er] + 31 trp-L[er] + 84 tyr-L[er] + 335 val-L[er] + 17556 atp[er] + 13167 h2o[er] --> Perlecan_Core_Protein[er] + 17556 adp[er] + 17556 pi[er] + 17556 h[er]

T_Core_Perlecan_ER_Golgi	Perlecan_Core_Protein[er] --> Perlecan_Core_Protein[g]	ER [er] -> Golgi [g]
R_HS_Elongation_on_Perlecan	Perlecan_Core_Protein[g] + 100 udpglcnac[g] + 100 udpglca[g] --> HS_Chain_100_on_Perlecan_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_HS_Sulfation_on_Perlecan	HS_Chain_100_on_Perlecan_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_HS_Proteoglycan_on_Perlecan[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Perlecan proteoglycan into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_HS_on_Perlecan_Packaging	Fully_Assembled_HS_Proteoglycan_on_Perlecan[g] --> Packaged_HS_on_Perlecan[v]	Golgi [g] -> Vesicle [v]
R_HS_on_Perlecan_Vesicle_Activation	Packaged_HS_on_Perlecan[v] + atp[c] + gtp[c] --> Activated_HS_on_Perlecan_Cargo[v] + adp[c] + pi[c] + gdp[c] + pi[c]	Vesicle [v], Cytosol [c]
R_HS_on_Perlecan_Secretion	Activated_HS_on_Perlecan_Cargo[v] --> Functional_Perlecan_HS[gl]	Vesicle [v] -> Glycocalyx Layer [gl]

(This block details the fate of the secreted Perlecan according to our superior hybrid model)

Reaction ID	Reaction Equation	Compartments Involved
R_Perlecan_HS_Degradation	Functional_Perlecan_HS[gl] + 9 h2o[gl] --> Truncated_Perlecan_Core[gl] + 10 Shed_HS_Fragment[gl] + 9 h[gl]	Glycocalyx [gl]
T_Shed_HS_Fragment_Diffusion	Shed_HS_Fragment[gl] --> Shed_HS_Fragment[e]	Glycocalyx [gl] -> Extracellular [e]
DM_Truncated_Perlecan	Truncated_Perlecan_Core[gl] -->	Glycocalyx [gl]
(Shared Sink)	Shed_HS_Fragment[e] -->	Extracellular [e]

Definition of New Metabolites for this Pathway

Metabolite ID	Compartment	Description
Perlecan_Core_Protein	[er], [g]	The unmodified 4390 amino acid protein backbone of Perlecan (UniProt P98160).
HS_Chain_100_on_Perlecan_Unsulfated	[g]	The Perlecan core with a 100-disaccharide unsulfated Heparan Sulfate chain.
Fully_Assembled_HS_Proteoglycan_on_Perlecan	[g]	The final product in the Golgi: Perlecan with a 100-unit sulfated HS chain.
Packaged_HS_on_Perlecan	[v]	The finished Perlecan-HS proteoglycan packaged

Activated_HS_on_Perlecan_Cargo	[v]	within a transport vesicle.
Functional_Perlecan_HS	[gl]	The Perlecan-HS vesicle cargo after being "powered up" for transport.
Truncated_Perlecan_Core	[gl]	The newly secreted Perlecan, transiently associated with the glycocalyx layer.
Shed_HS_Fragment	[gl], [e]	The Perlecan core protein remaining in the glycocalyx after its HS chains have been cleaved off.
		A soluble, 10-disaccharide HS fragment. It appears first in [gl] and then diffuses to [e].

4.4.2. Versican

4.4.2.1. Chondroitin Sulfate (CS) and Dermatan Sulfate (DS) Pathway

This subnetwork details the complete life cycle of a Versican proteoglycan, which is secreted into the pericellular environment.

(This block ensures all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GalNAc_Golgi	udpgal 鋹[c] <=> udpgal 鋹[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block details the synthesis of the unique Versican core protein and the parallel attachment of the three different CS GAG chains onto it)

Reaction ID	Reaction Equation	Compartment
R_Synth_Versican_Core	187 ala-L[er] + 115 arg-L[er] + 130 asn-L[er] + 253 asp-L[er] + 75 cys-L[er] + 130 gln-L[er] + 372 glu-L[er] + 219 gly[er] + 78 his-L[er] + 136 ile-L[er] + 242 leu-L[er] + 155 lys-L[er] + 37 met-L[er] + 109 phe-L[er] + 224 pro-L[er] + 286 ser-L[er] + 196 thr-L[er] + 41 trp-L[er] + 88 tyr-L[er] + 160 val-L[er] + 13568 atp[er] + 10176 h2o[er] --> Versican_Core_Protein[er] + 13568 adp[er] + 13568 pi[er] + 13568 h[er]	ER [er]
T_Versican_Core_ER_Golgi	Versican_Core_Protein[er] --> Versican_Core_Protein[g]	ER [er] -> Golgi [g]
R_Versican_	Versican_Core_Protein[g] + 100 udpgal 鋹[g] + 100 udpglca[g] -->	Golgi [g]

CS_Elongation	Versican_CS_Chain_100_Unsulfated[g] + 200 udp[g]	
R_Versican_CS_Sulfation_A	Versican_CS_Chain_100_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_Versican_CSA[g] + 100 pap[g]	Golgi [g]
R_Versican_CS_Sulfation_B	Versican_CS_Chain_100_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_Versican_CSB[g] + 100 pap[g]	Golgi [g]
R_Versican_CS_Sulfation_E	Versican_CS_Chain_100_Unsulfated[g] + 200 paps[g] --> Fully_Assembled_Versican_CSE[g] + 200 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Versican-CS proteoglycans into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_Versican_CSA_Packaging	Fully_Assembled_Versican_CSA[g] --> Packaged_Versican_CSA[v]	Golgi [g] -> Vesicle [v]
R_Versican_CSA_Vesicle_Activation	Packaged_Versican_CSA[v] + atp[c] + gtp[c] --> Activated_Versican_CSA_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Versican_CSA_Secretion	Activated_Versican_CSA_Cargo[v] --> Functional_Versican_CSA[gl]	Vesicle [v] -> Glycocalyx [gl]
R_Versican_CSB_Packaging	Fully_Assembled_Versican_CSB[g] --> Packaged_Versican_CSB[v]	Golgi [g] -> Vesicle [v]
R_Versican_CSB_Vesicle_Activation	Packaged_Versican_CSB[v] + atp[c] + gtp[c] --> Activated_Versican_CSB_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Versican_CSB_Secretion	Activated_Versican_CSB_Cargo[v] --> Functional_Versican_CSB[gl]	Vesicle [v] -> Glycocalyx [gl]
R_Versican_CSE_Packaging	Fully_Assembled_Versican_CSE[g] --> Packaged_Versican_CSE[v]	Golgi [g] -> Vesicle [v]
R_Versican_CSE_Vesicle_Activation	Packaged_Versican_CSE[v] + atp[c] + gtp[c] --> Activated_Versican_CSE_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Versican_CSE_Secretion	Activated_Versican_CSE_Cargo[v] --> Functional_Versican_CSE[gl]	Vesicle [v] -> Glycocalyx [gl]

(This block applies the final, corrected "N-to-1" degradation model and correctly clears only the shed fragments)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Versican_Shedding_A	Functional_Versican_CSA[gl] + 11 h2o[e] --> Truncated_Versican_Core_A[gl] + 12 Shed_CS_Fragment_A[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]

R_Versican_Shedding_B	Functional_Versican_CSB[gl] + 11 h2o[e] --> Truncated_Versican_Core_B[gl] + 12 Shed_CS_Fragment_B[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
R_Versican_Shedding_E	Functional_Versican_CSE[gl] + 11 h2o[e] --> Truncated_Versican_Core_E[gl] + 12 Shed_CS_Fragment_E[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Truncated_Versican	Truncated_Versican_Core_A[gl] → Truncated_Versican_Core_B[gl] → Truncated_Versican_Core_E[gl] -->	Glycocalyx [gl]
DM_Clearance_E	Shed_CS_Fragment_A[e] → Shed_CS_Fragment_B[e] → Shed_CS_Fragment_E[e] -->	Extracellular [e]

Definition of New Metabolites for the Versican Pathway

Metabolite ID	Compartment	Description
Versican_Core_Protein	[er], [g]	The unmodified 3393 amino acid protein backbone of Versican (UniProt P13611).
Versican_CS_Chain_100_Unsulfated	[g]	The Versican core with a 100-disaccharide unsulfated CS chain.
Fully_Assembled_Versican_CSA, ..._CSB, ..._CSE	[g]	The final, sulfated Versican proteoglycan products in the Golgi.
Packaged_Versican...	[v]	The finished Versican proteoglycans packaged within vesicles.
Activated_Versican...	[v]	The Versican vesicle cargo after being "powered up" for transport.
Functional_Versican_CSA, ..._CSB, ..._CSE	[gl]	The newly secreted Versican proteoglycans, transiently associated with the glycocalyx layer.
Truncated_Versican_Core_A, ..._B, ..._E	[gl]	The Versican core protein remaining in the glycocalyx after its CS chains have been cleaved off.
Shed_CS_Fragment_A, ..._B, ..._E	[e]	Soluble, ~8-disaccharide fragments of the respective CS types released into the extracellular space.

4.4.3. Biglycan

4.4.3.1. Chondroitin Sulfate (CS) and Dermatan Sulfate (DS) Pathway

This subnetwork details the complete life cycle of a Biglycan proteoglycan, which is secreted into the extracellular environment.

(This block is identical to the one used for the other CS pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]

T_UDP_GalNAc_Golgi	udpgal 𠄎[c] <=> udpgal 𠄎[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block details the synthesis of the unique Biglycan core protein and the attachment of CS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Biglycan_Core	11 ala-L[er] + 8 arg-L[er] + 8 asn-L[er] + 25 asp-L[er] + 4 cys-L[er] + 6 gln-L[er] + 23 glu-L[er] + 20 gly[er] + 7 his-L[er] + 13 ile-L[er] + 41 leu-L[er] + 15 lys-L[er] + 5 met-L[er] + 9 phe-L[er] + 21 pro-L[er] + 18 ser-L[er] + 11 thr-L[er] + 4 trp-L[er] + 10 tyr-L[er] + 21 val-L[er] + 1468 atp[er] + 1101 h2o[er] --> Biglycan_Core_Protein[er] + 1468 adp[er] + 1468 pi[er] + 1468 h[er]	ER [er]
T_Biglycan_Core_ER_Golgi	Biglycan_Core_Protein[er] --> Biglycan_Core_Protein[g]	ER [er] -> Golgi [g]
R_Biglycan_CS_Elongation	Biglycan_Core_Protein[g] + 100 udpgal 𠄎[g] + 100 udpglca[g] --> Biglycan_CS_Chain_100_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_Biglycan_CS_Sulfation_A_B	Biglycan_CS_Chain_100_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_Biglycan_CSA_B[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Biglycan-CS proteoglycan into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_Biglycan_CSA_B_Packaging	Fully_Assembled_Biglycan_CSA_B[g] --> Packaged_Biglycan_CSA_B[v]	Golgi [g] -> Vesicle [v]
R_Biglycan_CSA_B_Vesicle_Activation	Packaged_Biglycan_CSA_B[v] + atp[c] + gtp[c] --> Activated_Biglycan_CSA_B_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Biglycan_CSA_Secretion	Activated_Biglycan_CSA_B_Cargo[v] --> Functional_Biglycan_CSA[gl]	Vesicle [v] -> Glycocalyx [gl]
R_Biglycan_CSB_Secretion	Activated_Biglycan_CSA_B_Cargo[v] --> Functional_Biglycan_CSB[gl]	Vesicle [v] -> Glycocalyx [gl]

(This block applies the "N-to-1" (N=12) degradation model to the new Biglycan substrates)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Biglycan_Shearing_A	Functional_Biglycan_CSA[gl] + 11 h2o[e] --> Truncated_Biglycan_Core_A[gl] + 12	Glycocalyx [gl] -> [gl] + Extracellular [e]

	Shed_CS_Fragment_A[e] + 11 h[e]	
R_Biglycan_Shedd_B	Functional_Biglycan_CSB[gl] + 11 h2o[e] --> Truncated_Biglycan_Core_B[gl] + 12 Shed_CS_Fragment_B[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Truncated_Biglycan	Truncated_Biglycan_Core_A[gl] -- > Truncated_Biglycan_Core_B[gl] -->	Glycocalyx [gl]
(Shared Sinks)	Shed_CS_Fragment_A[e] -- > Shed_CS_Fragment_B[e] -->	Extracellular [e]

Definition of New Metabolites for the Biglycan Pathway

Metabolite ID	Compartment	Description
Biglycan_Core_Protein	[er], [g]	The unmodified 368 amino acid protein backbone of Biglycan (UniProt P21810).
Biglycan_CS_Chain_100_Unsulfated	[g]	The Biglycan core with a 100-disaccharide unsulfated CS chain.
Fully_Assembled_Biglycan_CSA_B	[g]	The final, sulfated Biglycan proteoglycan product in the Golgi.
Packaged_Biglycan_CSA_B	[v]	The finished Biglycan proteoglycan packaged within a vesicle.
Activated_Biglycan_CSA_B_Cargo	[v]	The Biglycan vesicle cargo after being "powered up" for transport.
Functional_Biglycan_CSA / ..._CSB	[gl]	The newly secreted Biglycan proteoglycans, transiently associated with the glycocalyx layer.
Truncated_Biglycan_Core_A / ..._B	[gl]	The Biglycan core protein remaining in the glycocalyx after its CS chains have been cleaved off.
(Shared Metabolites)	[e]	The pathway uses the same Shed_CS_Fragment_A and Shed_CS_Fragment_B metabolites as the other CS pathways.

4.4.4. Decorin

4.4.4.1. Chondroitin Sulfate (CS) and Dermatan Sulfate (DS) Pathway

This subnetwork details the complete life cycle of a Decorin proteoglycan, which is secreted into the extracellular environment.

(This block is identical to the one used for the other CS pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GalNAc_Golgi	udpgal 鈇[c] <=> udpgal 鈇[g]	Golgi [g]

T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block details the synthesis of the unique Decorin core protein and the attachment of CS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Decorin_Core	11 ala-L[er] + 9 arg-L[er] + 13 asn-L[er] + 20 asp-L[er] + 6 cys-L[er] + 9 gln-L[er] + 24 glu-L[er] + 23 gly[er] + 8 his-L[er] + 13 ile-L[er] + 44 leu-L[er] + 21 lys-L[er] + 4 met-L[er] + 8 phe-L[er] + 19 pro-L[er] + 24 ser-L[er] + 12 thr-L[er] + 2 trp-L[er] + 8 tyr-L[er] + 19 val-L[er] + 1432 atp[er] + 1074 h2o[er] --> Decorin_Core_Protein[er] + 1432 adp[er] + 1432 pi[er] + 1432 h[er]	ER [er]
T_Decorin_Core_ER_Golgi	Decorin_Core_Protein[er] --> Decorin_Core_Protein[g]	ER [er] -> Golgi [g]
R_Decorin_CS_Elongation	Decorin_Core_Protein[g] + 100 udpgal [g] + 100 udpglca[g] --> Decorin_CS_Chain_100_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_Decorin_CS_Sulfation_A_B	Decorin_CS_Chain_100_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_Decorin_CSA_B[g] + 100 pap[g]	Golgi [g]
T_Waste_UDP_P_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Decorin-CS proteoglycan into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_Decorin_CSA_B_Packaging	Fully_Assembled_Decorin_CSA_B[g] --> Packaged_Decorin_CSA_B[v]	Golgi [g] -> Vesicle [v]
R_Decorin_CSA_B_Vesicle_Activation	Packaged_Decorin_CSA_B[v] + atp[c] + gtp[c] --> Activated_Decorin_CSA_B_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Decorin_CSA_Secretion	Activated_Decorin_CSA_B_Cargo[v] --> Functional_Decorin_CSA[gl]	Vesicle [v] -> Glycocalyx [gl]
R_Decorin_CSB_Secretion	Activated_Decorin_CSA_B_Cargo[v] --> Functional_Decorin_CSB[gl]	Vesicle [v] -> Glycocalyx [gl]

(This block applies the "N-to-1" (N=12) degradation model to the new Decorin substrates)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Decorin_Shedding_A	Functional_Decorin_CSA[gl] + 11 h2o[e] --> Truncated_Decorin_Core_A[gl] + 12 Shed_CS_Fragment_A[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
R_Decorin_Shed	Functional_Decorin_CSB[gl] + 11 h2o[e] -->	Glycocalyx [gl] -> [gl] +

ding_B	Truncated_Decorin_Core_B[gl] + 12 Shed_CS_Fragment_B[e] + 11 h[e]	Extracellular [e]
DM_Truncated_Decorin	Truncated_Decorin_Core_A[gl] -- > Truncated_Decorin_Core_B[gl] -->	Glycocalyx [gl]
(Shared Sinks)	Shed_CS_Fragment_A[e] -- > Shed_CS_Fragment_B[e] -->	Extracellular [e]

Definition of New Metabolites for the Decorin Pathway

Metabolite ID	Compartment	Description
Decorin_Core_Protein	[er], [g]	The unmodified 359 amino acid protein backbone of Decorin (UniProt P07585).
Decorin_CS_Chain_100_Unsulfated	[g]	The Decorin core with a 100-disaccharide unsulfated CS chain.
Fully_Assembled_Decorin_CSA_B	[g]	The final, sulfated Decorin proteoglycan product in the Golgi.
Packaged_Decorin_CSA_B	[v]	The finished Decorin proteoglycan packaged within a vesicle.
Activated_Decorin_CSA_B_Cargo	[v]	The Decorin vesicle cargo after being "powered up" for transport.
Functional_Decorin_CSA / ..._CSB	[gl]	The newly secreted Decorin proteoglycans, transiently associated with the glycocalyx layer.
Truncated_Decorin_Core_A / ..._B	[gl]	The Decorin core protein remaining in the glycocalyx after its CS chains have been cleaved off.
(Shared Metabolites)	[e]	The pathway uses the same Shed_CS_Fragment_A and Shed_CS_Fragment_B metabolites as the other CS pathways.

4.3.5. Lumican

4.3.5.1. Keratan Sulfate (KS) Pathway

This subnetwork details the complete life cycle of a Lumican proteoglycan, decorated with representative Keratan Sulfate GAG chains, which is secreted into the extracellular environment.

(This block now includes the NEW transporters for the UDP-sugar precursors required for KS synthesis)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_Gal_Golgi (NEW)	udpgal[c] <=> udpgal[g]	Golgi [g]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]

T_H2O_Golgi	$\text{h2o}[\text{c}] \rightleftharpoons \text{h2o}[\text{g}]$	Golgi [g]
T_H_E	$\text{h}[\text{e}] \rightleftharpoons \text{h}[\text{c}]$	Extracellular [e]
T_H2O_E	$\text{h2o}[\text{e}] \rightleftharpoons \text{h}[\text{c}]$	Extracellular [e]

(This block details the synthesis of the unique Lumican core protein and the attachment of the new KS GAG chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Lumican_Core	$8 \text{ ala-L}[\text{er}] + 4 \text{ arg-L}[\text{er}] + 16 \text{ asn-L}[\text{er}] + 18 \text{ asp-L}[\text{er}] + 4 \text{ cys-L}[\text{er}] + 4 \text{ gln-L}[\text{er}] + 20 \text{ glu-L}[\text{er}] + 13 \text{ gly}[\text{er}] + 5 \text{ his-L}[\text{er}] + 12 \text{ ile-L}[\text{er}] + 43 \text{ leu-L}[\text{er}] + 13 \text{ lys-L}[\text{er}] + 6 \text{ met-L}[\text{er}] + 7 \text{ phe-L}[\text{er}] + 14 \text{ pro-L}[\text{er}] + 26 \text{ ser-L}[\text{er}] + 10 \text{ thr-L}[\text{er}] + 2 \text{ trp-L}[\text{er}] + 10 \text{ tyr-L}[\text{er}] + 23 \text{ val-L}[\text{er}] + 1348 \text{ atp}[\text{er}] + 1011 \text{ h2o}[\text{er}] \rightarrow \text{Lumican_Core_Protein}[\text{er}] + 1348 \text{ adp}[\text{er}] + 1348 \text{ pi}[\text{er}] + 1348 \text{ h}[\text{er}]$	ER [er]
T_Lumican_Core_ER_Golgi	$\text{Lumican_Core_Protein}[\text{er}] \rightarrow \text{Lumican_Core_Protein}[\text{g}]$	ER [er] -> Golgi [g]
R_Lumican_KS_Elongation	$\text{Lumican_Core_Protein}[\text{g}] + 25 \text{ udpgal}[\text{g}] + 25 \text{ udpglcnac}[\text{g}] \rightarrow \text{Lumican_KS_Chain_25_Unsulfated}[\text{g}] + 50 \text{ udp}[\text{g}]$	Golgi [g]
R_Lumican_KS_Sulfation	$\text{Lumican_KS_Chain_25_Unsulfated}[\text{g}] + 25 \text{ paps}[\text{g}] \rightarrow \text{Fully_Assembled_Lumican_KS}[\text{g}] + 25 \text{ pap}[\text{g}]$	Golgi [g]
T_Waste_UDP_out	$\text{udp}[\text{g}] \rightleftharpoons \text{udp}[\text{c}]$	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	$\text{pap}[\text{g}] \rightleftharpoons \text{pap}[\text{c}]$	Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Lumican-KS proteoglycan into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_Lumican_KS_Packaging	$\text{Fully_Assembled_Lumican_KS}[\text{g}] \rightarrow \text{Packaged_Lumican_KS}[\text{v}]$	Golgi [g] -> Vesicle [v]
R_Lumican_KS_Vesicle_Activation	$\text{Packaged_Lumican_KS}[\text{v}] + \text{atp}[\text{c}] + \text{gtp}[\text{c}] \rightarrow \text{Activated_Lumican_KS_Cargo}[\text{v}] + \text{adp}[\text{c}] + \text{gdp}[\text{c}] + 2 \text{ pi}[\text{c}]$	Vesicle [v], Cytosol [c]
R_Lumican_KS_Secretion	$\text{Activated_Lumican_KS_Cargo}[\text{v}] \rightarrow \text{Functional_Lumican_KS}[\text{gl}]$	Vesicle [v] -> Glycocalyx [gl]

(This block applies the "N-to-1" (N=5) degradation model for Keratan Sulfate)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Lumican_Shedding	$\text{Functional_Lumican_KS}[\text{gl}] + 4 \text{ h2o}[\text{e}] \rightarrow \text{Truncated_Lumican_Core}[\text{gl}] + 5 \text{ Shed_KS_Fragment}[\text{e}] + 4 \text{ h}[\text{e}]$	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Truncated_Lumican	$\text{Truncated_Lumican_Core}[\text{gl}] \rightarrow$	Glycocalyx [gl]
DM_Shed_KS_Fragment	$\text{Shed_KS_Fragment}[\text{e}] \rightarrow$	Extracellular [e]

Definition of New Metabolites for the Lumican Pathway

Metabolite ID	Compartment	Description
Lumican_Core_Protein	[er], [g]	The unmodified 338 amino acid protein backbone of Lumican (UniProt P51884).
Lumican_KS_Chain_25_Unsulfated	[g]	The Lumican core with a 25-disaccharide unsulfated Keratan Sulfate chain.
Fully_Assembled_Lumican_KS	[g]	The final, sulfated Lumican proteoglycan product in the Golgi.
Packaged_Lumican_KS	[v]	The finished Lumican proteoglycan packaged within a vesicle.
Activated_Lumican_KS_Cargo	[v]	The Lumican vesicle cargo after being "powered up" for transport.
Functional_Lumican_KS	[gl]	The newly secreted Lumican proteoglycan, transiently associated with the glycocalyx layer.
Truncated_Lumican_Core	[gl]	The Lumican core protein remaining in the glycocalyx after its KS chains have been cleaved off.
Shed_KS_Fragment	[e]	A soluble, 5-disaccharide Keratan Sulfate fragment released after degradation.

4.4.6. Fibromodulin

4.4.6.1. Keratan Sulfate (KS) Pathway

This subnetwork details the complete life cycle of a Fibromodulin proteoglycan, decorated with representative Keratan Sulfate GAG chains, which is secreted into the extracellular environment.

(This block is identical to the one used for the Lumican-KS pathway, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_Gal_Golgi	udpgal[c] <=> udpgal[g]	Golgi [g]
T_UDP_GlcNAc_Golgi	udpglcnac[c] <=> udpglcnac[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block details the synthesis of the unique Fibromodulin core protein and the attachment of the KS GAG chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Fibrom	17 ala-L[er] + 10 arg-L[er] + 13 asn-L[er] + 21 asp-L[er] + 4	ER [er]

odulin_Core	cys-L[er] + 6 gln-L[er] + 23 glu-L[er] + 16 gly[er] + 5 his-L[er] + 10 ile-L[er] + 48 leu-L[er] + 12 lys-L[er] + 5 met-L[er] + 9 phe-L[er] + 24 pro-L[er] + 28 ser-L[er] + 12 thr-L[er] + 4 trp-L[er] + 14 tyr-L[er] + 21 val-L[er] + 1500 atp[er] + 1125 h2o[er] --> Fibromodulin_Core_Protein[er] + 1500 adp[er] + 1500 pi[er] + 1500 h[er]	
T_Fibromodulin_Core_ER_Golgi	Fibromodulin_Core_Protein[er] --> Fibromodulin_Core_Protein[g]	ER [er] -> Golgi [g]
R_Fibromodulin_KS_Elongation	Fibromodulin_Core_Protein[g] + 25 udpgal[g] + 25 udpglcnac[g] --> Fibromodulin_KS_Chain_25_Unsulfated[g] + 50 udp[g]	Golgi [g]
R_Fibromodulin_KS_Sulfation	Fibromodulin_KS_Chain_25_Unsulfated[g] + 25 paps[g] --> Fully_Assembled_Fibromodulin_KS[g] + 25 pap[g]	Golgi [g]
T_Waste_UDP_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]
<i>(This block details the secretion of the finished Fibromodulin-KS proteoglycan into the pericellular space)</i>		

Reaction ID	Reaction Equation	Compartments Involved
R_Fibromodulin_KS_Packaging	Fully_Assembled_Fibromodulin_KS[g] --> Packaged_Fibromodulin_KS[v]	Golgi [g] -> Vesicle [v]
R_Fibromodulin_KS_Vesicle_Activation	Packaged_Fibromodulin_KS[v] + atp[c] + gtp[c] --> Activated_Fibromodulin_KS_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Fibromodulin_KS_Secretion	Activated_Fibromodulin_KS_Cargo[v] --> Functional_Fibromodulin_KS[gl]	Vesicle [v] -> Glycocalyx [gl]

(This block applies the "N-to-1" (N=5) degradation model for Keratan Sulfate)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Fibromodulin_Shedding	Functional_Fibromodulin_KS[gl] + 4 h2o[e] --> Truncated_Fibromodulin_Core[gl] + 5 Shed_KS_Fragment[e] + 4 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Truncated_Fibromodulin (Shared Sink)	Truncated_Fibromodulin_Core[gl] --> Shed_KS_Fragment[e] -->	Glycocalyx [gl] Extracellular [e]

Definition of New Metabolites for the Fibromodulin Pathway

Metabolite ID	Compartment	Description
Fibromodulin_Core_Protein	[er], [g]	The unmodified 376 amino acid protein backbone of Fibromodulin (UniProt Q06828).
Fibromodulin_KS_Chain_25_Unsulfated	[g]	The Fibromodulin core with a 25-disaccharide unsulfated Keratan Sulfate chain.
Fully_Assembled_Fibromodulin_KS	[g]	The final, sulfated Fibromodulin proteoglycan product in the Golgi.
Packaged_Fibromodulin_KS	[v]	The finished Fibromodulin proteoglycan packaged within a vesicle.

Activated_Fibromodulin_K S_Cargo	[v]	The Fibromodulin vesicle cargo after being "powered up" for transport.
Functional_Fibromodulin_KS	[gl]	The newly secreted Fibromodulin proteoglycan, transiently associated with the glycocalyx layer.
Truncated_Fibromodulin_Core	[gl]	The Fibromodulin core protein remaining in the glycocalyx after its KS chains have been cleaved off.
(Shared Metabolite)	[e]	The pathway uses the same Shed_KS_Fragment metabolite as the Lumican pathway.

4.4.7. Serglycin

4.4.7.1. Chondroitin Sulfate (CS) and Dermatan Sulfate (DS) Pathway

This subnetwork details the complete life cycle of a Serglycin proteoglycan, which is secreted into the extracellular environment.

(This block is identical to the one used for the other CS pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GalNAc_Golgi	udpgal 鈇[c] <=> udpgal 鈇[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block details the synthesis of the unique Serglycin core protein and the attachment of CS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Serglycin_Core	7 ala-L[er] + 6 arg-L[er] + 6 asn-L[er] + 10 asp-L[er] + 3 cys-L[er] + 6 gln-L[er] + 12 glu-L[er] + 15 gly[er] + 3 his-L[er] + 3 ile-L[er] + 20 leu-L[er] + 6 lys-L[er] + 4 met-L[er] + 6 phe-L[er] + 9 pro-L[er] + 15 ser-L[er] + 4 thr-L[er] + 2 trp-L[er] + 3 tyr-L[er] + 8 val-L[er] + 628 atp[er] + 471 h2o[er] --> Serglycin_Core_Protein[er] + 628 adp[er] + 628 pi[er] + 628 h[er]	ER [er]
T_Serglycin_Core_ER_Golgi	Serglycin_Core_Protein[er] --> Serglycin_Core_Protein[g]	ER [er] -> Golgi [g]
R_Serglycin_CS_Elongation	Serglycin_Core_Protein[g] + 100 udpgal 鈇[g] + 100 udpglca[g] --> Serglycin_CS_Chain_100_Unsulfated[g] + 200 udp[g]	Golgi [g]

R_Serglycin_CS_Sulfation_A_B	Serglycin_CS_Chain_100_Unsulfated[g] + 100 paps[g] --> Fully_Assembled_Serglycin_CSA_B[g] + 100 pap[g]	Golgi [g]
T_Waste_UD_P_out	udp[g] <=> udp[c]	Golgi [g] -> Cytosol [c]
T_Waste_PAP_out	pap[g] <=> pap[c]	Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Serglycin-CS proteoglycan into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_Serglycin_CSA_B_Packaging	Fully_Assembled_Serglycin_CSA_B[g] --> Packaged_Serglycin_CSA_B[v]	Golgi [g] -> Vesicle [v]
R_Serglycin_CSA_B_Vesicle_Activation	Packaged_Serglycin_CSA_B[v] + atp[c] + gtp[c] --> Activated_Serglycin_CSA_B_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Serglycin_CSA_Secretion	Activated_Serglycin_CSA_B_Cargo[v] --> Functional_Serglycin_CSA[gl]	Vesicle [v] -> Glycocalyx [gl]
R_Serglycin_CSB_Secretion	Activated_Serglycin_CSA_B_Cargo[v] --> Functional_Serglycin_CSB[gl]	Vesicle [v] -> Glycocalyx [gl]

(This block applies the "N-to-1" (N=12) degradation model to the new Serglycin substrates)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Serglycin_Shedding_A	Functional_Serglycin_CSA[gl] + 11 h2o[e] --> Truncated_Serglycin_Core_A[gl] + 12 Shed_CS_Fragment_A[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
R_Serglycin_Shedding_B	Functional_Serglycin_CSB[gl] + 11 h2o[e] --> Truncated_Serglycin_Core_B[gl] + 12 Shed_CS_Fragment_B[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Truncated_Serglycin	Truncated_Serglycin_Core_A[gl] --> Truncated_Serglycin_Core_B[gl] --> Shed_CS_Fragment_A[e] --> Shed_CS_Fragment_B[e] -->	Glycocalyx [gl] Extracellular [e]

Definition of New Metabolites for the Serglycin Pathway

Metabolite ID	Compartment	Description
Serglycin_Core_Protein	[er], [g]	The unmodified 158 amino acid protein backbone of Serglycin (UniProt P10124).
Serglycin_CS_Chain_100_Unsulfated	[g]	The Serglycin core with a 100-disaccharide unsulfated CS chain.
Fully_Assembled_Serglycin_CSA_B	[g]	The final, sulfated Serglycin proteoglycan product in the Golgi.
Packaged_Serglycin_CSA_B	[v]	The finished Serglycin proteoglycan packaged within a vesicle.
Activated_Serglycin_CSA_B_Cargo	[v]	The Serglycin vesicle cargo after being "powered up" for transport.
Functional_Serglycin_CSA	[gl]	The newly secreted Serglycin proteoglycans, transiently

CSA / ..._CSB		associated with the glycocalyx layer.
Truncated_Serglycin_Core_A / ..._B	[gl]	The Serglycin core protein remaining in the glycocalyx after its CS chains have been cleaved off.
(Shared Metabolites)	[e]	The pathway uses the same Shed_CS_Fragment_A and Shed_CS_Fragment_B metabolites as the other CS pathways.

4.3.8. Aggrecan

4.3.8.1. Chondroitin Sulfate (CS) and Dermatan Sulfate (DS) Pathway

This subnetwork details the complete life cycle of an Aggrecan proteoglycan, which is secreted into the extracellular environment.

(This block is identical to the one used for the other CS pathways, ensuring all necessary precursors are transported to the ER and Golgi)

Reaction ID	Reaction Equation	Target Compartment
T_Ala_ER to T_Val_ER	(amino_acid)[c] <=> (amino_acid)[er] (20 reactions)	Endoplasmic Reticulum [er]
T_ATP_ER	atp[c] <=> atp[er]	Endoplasmic Reticulum [er]
T_H2O_ER	h2o[c] <=> h2o[er]	Endoplasmic Reticulum [er]
T_UDP_GalNAc_Golgi	udpgal 𐄂[c] <=> udpgal 𐄂[g]	Golgi [g]
T_UDP_GlcA_Golgi	udpglca[c] <=> udpglca[g]	Golgi [g]
T_PAPS_Golgi	paps[c] <=> paps[g]	Golgi [g]
T_H2O_Golgi	h2o[c] <=> h2o[g]	Golgi [g]
T_H_E	h[e] <=> h[c]	Extracellular [e]
T_H2O_E	h2o[e] <=> h[c]	Extracellular [e]

(This block details the synthesis of the unique Aggrecan core protein and the attachment of CS chains)

Reaction ID	Reaction Equation	Compartment
R_Synth_Aggrecan_Core	110 ala-L[er] + 83 arg-L[er] + 93 asn-L[er] + 120 asp-L[er] + 59 cys-L[er] + 109 gln-L[er] + 288 glu-L[er] + 203 gly[er] + 54 his-L[er] + 83 ile-L[er] + 154 leu-L[er] + 91 lys-L[er] + 24 met-L[er] + 76 phe-L[er] + 205 pro-L[er] + 258 ser-L[er] + 189 thr-L[er] + 19 trp-L[er] + 46 tyr-L[er] + 153 val-L[er] + 9656 atp[er] + 7242 h2o[er] --> Aggrecan_Core_Protein[er] + 9656 adp[er] + 9656 pi[er] + 9656 h[er]	ER [er]
T_Aggrecan_Core_ER_Golgi	Aggrecan_Core_Protein[er] --> Aggrecan_Core_Protein[g]	ER [er] -> Golgi [g]
R_Aggrecan_CS_Elongation	Aggrecan_Core_Protein[g] + 100 udpgal 𐄂[g] + 100 udpglca[g] --> Aggrecan_CS_Chain_100_Unsulfated[g] + 200 udp[g]	Golgi [g]
R_Aggrecan_CS_Elongation	Aggrecan_CS_Chain_100_Unsulfated[g] + 100 paps[g] -->	Golgi [g]

CS_Sulfation_A_B Fully_Assembled_Aggrecan_CSA_B[g] + 100 pap[g]

T_Waste_UD P_out udp[g] <=> udp[c] Golgi [g] -> Cytosol [c]

T_Waste_PAP_out pap[g] <=> pap[c] Golgi [g] -> Cytosol [c]

(This block details the secretion of the finished Aggrecan-CS proteoglycan into the pericellular space)

Reaction ID	Reaction Equation	Compartments Involved
R_Aggrecan_CSA_B_Packaging	Fully_Assembled_Aggrecan_CSA_B[g] --> Packaged_Aggrecan_CSA_B[v]	Golgi [g] -> Vesicle [v]
R_Aggrecan_CSA_B_Vesicle_Activation	Packaged_Aggrecan_CSA_B[v] + atp[c] + gtp[c] --> Activated_Aggrecan_CSA_B_Cargo[v] + adp[c] + gdp[c] + 2 pi[c]	Vesicle [v], Cytosol [c]
R_Aggrecan_CSA_Secretion	Activated_Aggrecan_CSA_B_Cargo[v] --> Functional_Aggrecan_CSA[gl]	Vesicle [v] -> Glycocalyx [gl]
R_Aggrecan_CSB_Secretion	Activated_Aggrecan_CSA_B_Cargo[v] --> Functional_Aggrecan_CSB[gl]	Vesicle [v] -> Glycocalyx [gl]

(This block applies the "N-to-1" (N=12) degradation model to the new Aggrecan substrates)

Reaction ID (Model)	Reaction Equation	Compartments Involved
R_Aggrecan_Shedding_A	Functional_Aggrecan_CSA[gl] + 11 h2o[e] --> Truncated_Aggrecan_Core_A[gl] + 12 Shed_CS_Fragment_A[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
R_Aggrecan_Shedding_B	Functional_Aggrecan_CSB[gl] + 11 h2o[e] --> Truncated_Aggrecan_Core_B[gl] + 12 Shed_CS_Fragment_B[e] + 11 h[e]	Glycocalyx [gl] -> [gl] + Extracellular [e]
DM_Truncated_Aggrecan	Truncated_Aggrecan_Core_A[gl] --> Truncated_Aggrecan_Core_B[gl] -->	Glycocalyx [gl]
(Shared Sinks)	Shed_CS_Fragment_A[e] --> Shed_CS_Fragment_B[e] -->	Extracellular [e]

Definition of New Metabolites for the Aggrecan Pathway

Metabolite ID	Compartment	Description
Aggrecan_Core_Protein	[er], [g]	The unmodified 2415 amino acid protein backbone of Aggrecan (UniProt P16112).
Aggrecan_CS_Chain_100_Unsulfated	[g]	The Aggrecan core with a 100-disaccharide unsulfated CS chain.
Fully_Assembled_Aggrecan_CSA_B	[g]	The final, sulfated Aggrecan proteoglycan product in the Golgi.
Packaged_Aggrecan_CSA_B	[v]	The finished Aggrecan proteoglycan packaged within a vesicle.
Activated_Aggrecan_CSA_B_Cargo	[v]	The Aggrecan vesicle cargo after being "powered up" for transport.
Functional_Aggrecan_CSA / ..._CSB	[gl]	The newly secreted Aggrecan proteoglycans, transiently associated with the glycocalyx layer.

Truncated_AggreCAN_C
ore_A / ..._B [gl]

(Shared Metabolites) [e]

The AggreCAN core protein remaining in the glycocalyx after its CS chains have been cleaved off.

The pathway uses the same Shed_CS_Fragment_A and Shed_CS_Fragment_B metabolites as the other CS pathways.

4.5. Hyaluronan (HA) Pathway

4.5.1. Design Considerations

Hyaluronan is unique among GAGs as it is not attached to a core protein and is synthesized directly at the inner leaflet of the plasma membrane by Hyaluronan Synthase (HAS) enzymes. The growing polymer is extruded directly into the extracellular space. This pathway correctly bypasses the ER-Golgi and vesicular transport machinery used by proteoglycans.

Our model captures the critical pathological conversion of structural, anti-inflammatory High-Molecular-Weight (HMW) HA to bioactive, pro-inflammatory Low-Molecular-Weight (LMW) fragments. The stoichiometry for this "N-to-1" fragmentation is based on literature-derived values.

- **Justification for Native HMW-HA Chain Length (n = 2500 disaccharides):**
 - **Source:** Fraser, J. R., et al. (1997). Hyaluronan: its nature, distribution, functions and turnover. *Journal of internal medicine*, 242(1), 27-33.
 - **Rationale:** This classic review establishes that newly synthesized HA is a massive polymer, often exceeding 1,000 kDa. A representative average of **1,000 kDa** corresponds to **2500 disaccharide units** (1000 kDa / 0.4 kDa per unit), providing the value for 'n'.
- **Justification for Pro-inflammatory LMW-HA Fragment Size (~100 disaccharides):**
 - **Source:** Termeer, C., et al. (2002). Oligosaccharides of hyaluronan activate dendritic cells via toll-like receptor 4. *The Journal of experimental medicine*, 195(1), 99-111.
 - **Rationale:** This seminal paper demonstrates that the pro-inflammatory activity of HA fragments (acting as DAMPs via TLR4) is potent in the 20-50 kDa range. A representative **40 kDa** fragment corresponds to **100 disaccharide units**.
- **Final Calculation for N:** $N = 2500 / 100 = 25$.

4.4.2. Definitive Unabridged Reaction Set for the HA Pathway

(This block is simplified as the HAS enzyme draws directly from the cytosolic pool. The necessary UDP-sugars are assumed to be available from the main cellular model.)

Reaction ID Reaction Equation Target Compartment
T_H2O_GL $\text{h2o}[e] \rightleftharpoons \text{h2o}[gl]$ Glycocalyx Layer [gl]

(This single, irreversible reaction represents the entire synthesis and extrusion pathway, spanning the cytosolic and glycocalyx compartments.)

Reaction ID	Reaction Equation	Compartment(s)
R_Hyaluronan_Synt	$2500 \text{ udpglca}[c] + 2500 \text{ udpglcnac}[c] \rightarrow$	$[c] \rightarrow [gl]$

hesis Functional_HMW_Hyaluronan[gl] + 5000 udp[c]
 (This block details the one-way transformation from the HMW form to LMW fragments and their final clearance.)

Reaction ID	Reaction Equation	Compartments Involved
R_Hyaluronan_Fragmentation	Functional_HMW_Hyaluronan[gl] + 24 h2o[gl] --> 25 Shed_LMW_Hyaluronan_Fragment[e] + 24 h[gl]	Glycocalyx [gl] -> Extracellular [e]
DM_LMW_Fragment	Shed_LMW_Hyaluronan_Fragment[e] -->	Extracellular [e]

Metabolite ID	Compartment	Description
Functional_HMW_Hyaluronan	[gl]	The final, full-length (2500-disaccharide), anti-inflammatory Hyaluronan polymer woven into the glycocalyx layer.
Shed_LMW_Hyaluronan_Fragment	[e]	A soluble, 100-disaccharide, pro-inflammatory fragment released into the extracellular space after enzymatic or ROS-mediated cleavage.

5. Conclusion: A Comprehensive and Extensible Modeling Framework

This report documents the systematic development of a comprehensive, multi-pathway metabolic subnetwork for the endothelial glycocalyx. By integrating detailed biochemical knowledge with user-provided reaction data, we have constructed a model that accurately represents the synthesis and pathological turnover of the four major glycocalyx components. This framework is now a robust and powerful tool for simulating the complex metabolic shifts and compositional changes that occur during SHINE.

6. References and Data Sources

- Reitsma, S., et al. (2007).** The endothelial glycocalyx: composition, functions, and visualization. *Pflügers Archiv-European Journal of Physiology*, 454(3), 345–359.
 - Relevance:** Provided the foundational overview of the glycocalyx structure, its role in mechanotransduction and barrier function, and established the dominance of Heparan Sulfate.
- Chelazzi, C., et al. (2015).** The endothelial glycocalyx: a key player in the pathogenesis of sepsis. *Journal of Critical Care*, 30(6), 1317-1325.
 - Relevance:** Justified the focus on SHINE/sepsis as a state of massive glycocalyx shedding and provided evidence for the pro-inflammatory compositional shift.
- Habuchi, H., et al. (2008).** The role of heparan sulfate 6-O-sulfotransferases in the context of the functional sulfation code. *Glycoconjugate journal*, 25(1), 5-13.
 - Relevance:** Provided the justification for modeling *in situ* remodeling by describing the function of extracellular sulfatases (Sulfs) in editing the GAG sulfation code on the cell surface.

4. **Iozzo, R. V., & Schaefer, L. (2015).** Proteoglycan form and function: A comprehensive nomenclature of proteoglycans. *Matrix Biology*, 42, 11–55.
 - **Relevance:** This comprehensive review was the primary source for the hierarchical structure of the glycocalyx, the classification of core proteins (Syndecans, Versican), and the average length of GAG chains.
5. **Fraser, J. R., Laurent, T. C., & Laurent, U. B. (1997).** Hyaluronan: its nature, distribution, functions and turnover. *Journal of internal medicine*, 242(1), 27-33.
 - **Relevance:** This classic review was the source for the unique, non-proteoglycan nature of Hyaluronan and the basis for its average native molecular weight (1000+ kDa), which was used to calculate the stoichiometry for 'n'.
6. **Ramani, V. C., et al. (2012).** The role of syndecan-1 in the pathogenesis of acute lung injury. *American journal of respiratory and critical care medicine*, 186(11), 1087-1096.
 - **Relevance:** Provided direct evidence for the dual-pathology model by describing the roles of both proteolytic shedding (via MMPs) and heparanase-mediated degradation of Syndecan-1 in inflammatory lung injury.
7. **Vlodavsky, I., et al. (2007).** Mammalian heparanase: gene cloning, expression and function in health and disease. *Glycoconjugate journal*, 24(1), 43-55.
 - **Relevance:** This authoritative review was the source for the average size of bioactive, heparanase-generated HS fragments (5-10 kDa or ~10 disaccharides), which was used to calculate the stoichiometry for 'N'.
8. **Manon-Jensen, T., et al. (2016).** Proteoglycans in inflammation and tissue repair. *European journal of pharmacology*, 785, 40-54.
 - **Relevance:** Provided the justification for modeling Syndecan-1 as a hybrid proteoglycan, describing how the ratio of HS to CS chains on syndecans can be actively modulated during inflammatory responses.
9. **Wight, T. N. (2002).** Versican: a versatile extracellular matrix proteoglycan in cell biology. *Current opinion in cell biology*, 14(5), 617-623.
 - **Relevance:** Justified the creation of a separate pathway for Versican, describing its structure as a major CS-specific proteoglycan and its known upregulation during vascular inflammation and remodeling.
10. **Weigel, P. H., & DeAngelis, P. L. (2007).** Hyaluronan synthases: a decade-plus of novel twists and turns. *Journal of Biological Chemistry*, 282(51), 36777-36781.
 - **Relevance:** This review provided the definitive mechanism for HA synthesis, confirming its location at the plasma membrane and its direct extrusion into the extracellular space, justifying the unique topological design of our HA pathway.
11. **Termeer, C., et al. (2002).** Oligosaccharides of hyaluronan activate dendritic cells via toll-like receptor 4. *The Journal of experimental medicine*, 195(1), 99-111.
 - **Relevance:** This seminal paper was the source for the size of pro-inflammatory LMW-HA fragments (~40 kDa or ~100 disaccharides), which are potent activators of

innate immunity and were used to calculate the stoichiometry for 'N' in the HA fragmentation reaction.

12. **Thiele, I., & Palsson, B. Ø. (2010).** A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols*, 5(1), 93–121.

- **Relevance:** Provided the standard methodology for constructing all macromolecule synthesis reactions, including the calculation of the ATP cost of translation from primary protein sequences obtained from UniProt.